

Economic Inflection Points in Space Manufacturing: Monte Carlo Analysis of In-Situ Resource Utilization vs. Earth Launch for Large-Scale Space Infrastructure

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Abstract

We compare Earth-launch and ISRU pathways for serial production of unpressurized structural modules via a parametric NPV cost model with phased capital, pathway-specific schedules, Wright learning curves with stochastic saturation, and a dynamic vitamin fraction. A 10,000-run Monte Carlo at $r = 5\%$ (canonical configuration: Table 9) finds crossover in 85% of draws (conditional median $\sim 4,300$; Kaplan-Meier median $\sim 5,350$; 95% CI: [4,200, 4,420]). Of converging runs, 22% are analytically permanent; the remaining 78% are functionally permanent (savings window width $\sim 196,000$ units). Conditional on the assumed priors and model structure, 95% of draws (95% CI: [94%, 95%]) place a 20,000-unit program within the ISRU savings window. Results are reported at $\sigma_{\text{ln}} = 0.70$ (megaproject reference class); a space-specific variant ($\sigma_{\text{ln}} = 1.0$) appears in Table 7.

ISRU capital K and Earth learning rate explain $\sim 83\%$ of variance. K alone explains 63% and dominates the crossover surface: at K median \$65B, crossover probability is 85%; at \$150B, 46% (Table 14). Model-form sensi-

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¹This research was conducted using an AI-assisted methodology. Claude (Anthropic), GPT (OpenAI), and Gemini (Google) were used for literature synthesis; editorial review and peer review simulation were conducted using all three models. The Monte Carlo simulation code was written and verified by the human author in Python using NumPy and SciPy. All quantitative results were produced by deterministic and stochastic simulation code; parameter ranges were informed by published literature and engineering analogy. No AI-generated numerical outputs were used without independent verification against the simulation code.

tivity (piecewise plateau vs. logistic saturation, Eq. 22) shifts the crossover by $\pm 1,500$ units without eliminating it; the comparison is deterministic, and only the piecewise plateau is integrated into the stochastic MC. Three factors prevent crossover: vitamin costs $> \sim \$50,000/\text{kg}$, discount rates $> \sim 20\%$, or $p_s \lesssim 70\%$. For revenue-generating infrastructure, deployment delay offsets cost savings above $\sim \$0.94\text{M}/\text{unit}/\text{yr}$ (Eq. 27), suggesting the ISRU advantage is strongest for non-revenue infrastructure.

Keywords: in-situ resource utilization, space manufacturing, launch cost, Monte Carlo simulation, learning curves, net present value, space economics

1. Introduction

Any program of large-scale space infrastructure construction—whether solar power satellites, orbital habitats, or distributed energy collection architectures—eventually reaches a decision point that no amount of launch cost reduction can fully avoid under Earth-supplied logistics: at what scale does it become cheaper to manufacture components in space rather than to build them on Earth and haul them up through the gravity well? At megascale, some fraction of each unit’s mass must still be Earth-sourced (hereafter “vitamin” components, §3.2.4). This question has been posed qualitatively since at least the 1970s [26, 27], and it has recently gained renewed urgency as launch costs decline precipitously [18], lunar ISRU programs advance under the Artemis architecture [30], and private ventures pursue asteroid resource extraction [14, 1].

The intuition is rooted in a cost asymmetry that launch cost reduction cannot fully eliminate. Reusable vehicles have brought per-kilogram costs down by roughly two orders of magnitude, from $\$20,000/\text{kg}$ to projections below $\$500/\text{kg}$ for Starship-class vehicles. But per-kilogram launch costs decline more slowly than manufacturing costs: even under aggressive launch learning ($\text{LR}_L = 0.90$), the propellant and operations component ($\sim \$200/\text{kg}$) constitutes an assumed operational asymptote under Earth-supplied logistics that the learnable component cannot breach. ISRU inverts this cost structure: large fixed capital costs up front, but each additional unit produced from local materials costs only energy and labor. The fixed costs divide across a growing denominator; marginal costs decline with experience.

Despite the intuitive appeal of this argument, a quantitative gap persists in the literature. Existing ISRU economic analyses are overwhelmingly

mission-specific: they evaluate the cost-effectiveness of producing lunar oxygen for propellant [30], extracting water ice for life support [39], or mining asteroidal platinum-group metals for terrestrial markets [31, 14]. We are not aware of prior work that combines schedule-aware NPV crossover analysis with systematic uncertainty characterization for generic manufactured products.

A second gap concerns the treatment of manufacturing learning. The Wright learning curve [37], well established in aerospace production economics [36], predicts that unit manufacturing cost decreases as a power function of cumulative production volume. This effect has been extensively documented for Earth-based aerospace systems [2, 24] but we are not aware of it being combined with NPV timing analysis in a comparative Earth-versus-ISRU cost model for generic structural units. The omission matters because the two pathways exhibit fundamentally different learning dynamics: Earth manufacturing benefits from learning in fabrication but exhibits limited learning in launch (baseline $LR_L = 0.97$; we sweep from 90% to 100% in §4.2), while ISRU benefits from learning in both extraction and fabrication, albeit from a higher initial cost base and with a slower initial ramp-up. The long-run consequences of this asymmetry are not obvious without a quantitative model.

Our analysis is primarily relevant to non-revenue infrastructure (habitats, depots, structural frameworks) where cost minimization is the decision criterion. For revenue-generating systems such as space solar power, the opportunity cost of ISRU’s deployment delay may dominate the cost savings (§5), making the Earth pathway preferred even above the cost crossover.

This paper makes three contributions. First, we present a parametric cost model with net present value discounting and pathway-specific delivery schedules that expresses the total cost of producing N units via the Earth-launch pathway and the ISRU pathway as explicit functions of production volume, with learning curves, an integrated production ramp-up schedule, ISRU mass penalty and transport costs, and financing costs applied to the appropriate components (§3). Second, we embed this model in a 10,000-run Monte Carlo framework with correlated sampling that propagates uncertainty in eighteen independent stochastic parameters (plus two derived)—including the ISRU operational cost floor, production rate, and transport duration—at each of several fixed discount rates, yielding probability distributions, convergence analysis, and global sensitivity rankings rather than point estimates (§4). Third, we propose a phased hybrid transition strategy

and discuss its implications for near-term space policy (§5).

2. Related Work

The economic case for ISRU has been examined primarily for mission-specific applications. Sanders and Larson [30] found lunar oxygen extraction cost-effective after approximately five crewed missions, but their model does not generalize to manufactured goods. Crawford [8] catalogued lunar mineralogical inventories; Cilliers et al. [7] examined regolith processing pathways; Kornuta et al. [21] developed a commercial lunar propellant architecture. The MOXIE experiment [15] provided empirical validation of in-situ oxygen production on Mars, though at sub-manufacturing scales. Sowers [32, 33] presented NPV-based business cases for lunar ice mining and cislunar economics. Metzger et al. [23] proposed a bootstrapping approach in which a seed factory progressively builds its own capacity. Asteroid mining NPV frameworks have been developed by Sonter [31], Elvis [14], and Andrews et al. [1]; Ishimatsu et al. [16] modeled Earth–Moon–Mars logistics as a multi-commodity network flow. These analyses focus on resource extraction rather than structural manufacturing. Lunar additive manufacturing demonstrations [35, 6] provide empirical grounding for the ISRU manufacturing pathway assumed here. O’Neill [26, 27] argued on energetic grounds for lunar material delivery to orbit, but lacked a parametric cost model and did not incorporate learning curves. The real options framework [10] extends NPV for irreversible investments under uncertainty; Saleh et al. [29] applied flexibility analysis to spacecraft, and de Weck et al. [9] demonstrated staged deployment value. Organizational forgetting [5, 34] has been documented in aerospace production lines.

Jones [18, 19, 20] documented a roughly two-orders-of-magnitude reduction in launch cost per kilogram over six decades. A critical distinction is between vehicle manufacturing cost (which follows a learning curve) and per-kilogram delivery cost (dominated by propellant and operations, largely independent of cumulative launches). Falcon 9 reuse economics [38] confirm that per-launch cost reductions arise from fixed-cost amortization at high flight rates, not production learning. Wertz [36] estimated launch vehicle production learning rates of 85–90%; the NASA Cost Estimating Handbook [25] provides standardized parametric methodologies that inform our parameter selection. Our model adopts a two-component launch cost: a propellant

operational asymptote plus a learnable component ($LR_L = 0.97$); we sweep LR_L from 0.90 to 1.00 in §4.2.

The Wright learning curve [37]—in which unit cost decreases as a power function of cumulative production—is well established in aerospace economics [11, 2, 24]; we formalize it in §3 (Eq. 2). Typical aerospace learning rates range from 0.80 to 0.95 [36]; Baumers et al. [4] documented 0.85–0.92 for metal additive manufacturing. We are not aware of prior work applying the Wright model to a comparative Earth-versus-ISRU manufacturing analysis with NPV timing.

3. Model Description

Decision problem. The primary comparison metric is the NPV of cumulative cost to deliver the first N units (Eq. 16), where each pathway delivers according to its own schedule. This answers: “Given a program requirement of N structural modules, which pathway has lower total present-value cost?” Section 5 extends this to a utility-maximizing framing (Eq. 27) that accounts for revenue per unit and the opportunity cost of ISRU’s deployment delay.

We model the total cost of producing N identical structural modules (hereafter “units”) via two pathways: an Earth-launch pathway and an ISRU pathway. Each unit has a reference mass $m = 1,850$ kg, representative of structural modules in the 1,000–5,000 kg class typical of solar collector or orbital habitat components, though the model is parameterizable to other mass ranges. We describe each pathway’s cost model, then the Monte Carlo framework used to propagate parameter uncertainty.

Reference orbit. Throughout this paper, “operational orbit” refers to geostationary Earth orbit (GEO), consistent with space solar power and communications relay architectures. The launch cost p_{launch} represents delivery from Earth surface to GEO; the transport cost $p_{\text{transport}}$ represents delivery from the lunar surface to GEO via low-energy trajectory (~ 6 km/s Δv). For other destinations (LEO, NRHO, cislunar), both cost terms would differ; the model is parameterizable to any orbit.

Cost basis normalization. All launch costs in this paper are quoted for delivery to GEO. Published Starship cost projections (\$200–500/kg) typically refer to LEO; delivery to GEO via an upper stage or electric tug adds a factor of ~ 2 – $3\times$, so \$500/kg to LEO corresponds to approximately \$1,000–1,500/kg

to GEO, consistent with our baseline of \$1,000/kg. The operational asymptote p_{fuel} is now sampled stochastically: $p_{\text{fuel}} \sim U[\$100, \$400]/\text{kg}$ (baseline \$200/kg). A bottom-up decomposition of the GEO delivery cost chain (propellant, LEO-to-GEO transfer, ground operations; Appendix Appendix D) yields a total floor of $\sim \$105\text{--}178/\text{kg}$, with \$200/kg as the central estimate; the $U[\$100, \$400]$ range brackets optimistic and conservative assumptions.

3.1. Earth-launch pathway

The cost of the n th unit produced on Earth and launched to the operational orbit is

$$C_{\text{Earth}}(n) = C_{\text{mfg}}(n) + C_{\text{launch}}(n), \quad (1)$$

where $C_{\text{mfg}}(n)$ is the terrestrial manufacturing cost and $C_{\text{launch}}(n)$ is the launch cost. Manufacturing cost follows a two-component model separating non-learnable material costs from learnable labor and overhead:

$$C_{\text{mfg}}(n) = C_{\text{mat}} + C_{\text{labor}}^{(1)} \cdot n^{b_E}, \quad b_E = \frac{\ln(\text{LR}_E)}{\ln 2}, \quad (2)$$

where $C_{\text{mat}} = \$1\text{M}$ is the non-learnable per-unit material cost (aerospace-grade aluminum alloy at $\sim \$540/\text{kg} \times 1,850 \text{ kg}$) and $C_{\text{labor}}^{(1)} = \74M is the first-unit recurring cost (labor, overhead, and tooling/NRE amortized across the first production lot), declining via the Wright curve with progress ratio LR_E ($= 0.85$: 15% cost reduction per doubling of cumulative output; note that LR closer to 1 means *slower* learning). Note on cost convention: $C_{\text{labor}}^{(1)}$ includes lot-amortized tooling ($\sim \$12\text{M}$) as is standard in aerospace production economics [36]: tooling is rebuilt or refurbished per production lot, and the resulting per-unit charge declines with cumulative output alongside labor learning. This differs from a facility-level NRE capex (which would be treated as a fixed charge outside the learning curve). The Wright curve therefore applies to the full recurring envelope, not solely to touch labor. The \$75M first-unit total reflects a spacecraft-class structural module with substantial assembly complexity; the build-up is: structural fabrication and welding ($\sim \$30\text{M}$), precision machining and finishing ($\sim \$15\text{M}$), quality assurance and testing ($\sim \$10\text{M}$), tooling amortization ($\sim \$12\text{M}$), and project management ($\sim \$8\text{M}$). For simpler commodity structures (beams, trusses), the first-unit cost would be lower ($\sim \$15\text{--}30\text{M}$); the archetype sensitivity (Table 2) tests this range. An optional manufacturing cost floor $C_{\text{mfg}}^{\text{floor}}$ can be applied; the

baseline uses $C_{\text{mfg}}^{\text{floor}} = 0$ (the material cost C_{mat} serves as the natural floor).

$$C_{\text{mfg}}(n) = \max\left(C_{\text{mat}} + C_{\text{labor}}^{(1)} \cdot n^{b_E}, C_{\text{mfg}}^{\text{floor}}\right). \quad (3)$$

As a reference case, launch cost can be treated as constant when the program's launch demand is a small fraction of total market demand:

$$C_{\text{launch}}(n) = m \cdot p_{\text{launch}}, \quad (4)$$

i.e., a constant $\$1,000/\text{kg} \times 1,850 \text{ kg} = \1.85M per unit. The baseline MC instead uses a two-component model with program-indexed learning:

$$C_{\text{launch}}^{\text{learn}}(n) = m \left[p_{\text{fuel}} + p_{\text{ops}} \cdot n^{b_L} \right], \quad b_L = \frac{\ln(\text{LR}_L)}{\ln 2}, \quad (5)$$

where $p_{\text{fuel}} = \$200/\text{kg}$ is an assumed operational asymptote under Earth-supplied propellant logistics (propellant plus minimal GEO delivery operations) and $p_{\text{ops}} = \$800/\text{kg}$ is the learnable component, declining at $\text{LR}_L = 0.97$ per doubling (see Table D.30).

In the Monte Carlo, p_{launch} is sampled from $U[500, 2000]$ and p_{fuel} from $U[100, 400]$. The learnable component is derived as $p_{\text{ops}} = \max(p_{\text{launch}} - p_{\text{fuel}}, 0)$, ensuring the total first-unit launch cost equals the sampled p_{launch} while the operational asymptote matches p_{fuel} .

For the deterministic baseline, the two formulations yield practically identical results: at baseline scale ($\sim 4,400$ units), the ops component under $\text{LR}_L = 0.97$ declines by less than 0.3%, a shift smaller than the search grid resolution (Table A.23). We sweep LR_L from 0.90 to 1.00 in §4.2; even at aggressive learning the crossover shifts by only +315 units (+7%), confirming that the model's conclusions are insensitive to the launch learning assumption.

Indexing convention. The learning index n counts cumulative program units. At the assumed program scale ($\sim 4,100$ – $10,000$ units), the program would constitute a substantial fraction of global launch demand, making program-indexed learning a reasonable proxy for market-indexed learning.

The cumulative cost of producing N units via the Earth pathway is therefore

$$\Sigma_{\text{Earth}}(N) = \sum_{n=1}^N \left[\max\left(C_{\text{mat}} + C_{\text{labor}}^{(1)} \cdot n^{b_E}, C_{\text{mfg}}^{\text{floor}}\right) + m \cdot p_{\text{launch}} \right]. \quad (6)$$

When the launch learning sensitivity variant (Eq. 5) is active, $m \cdot p_{\text{launch}}$ is replaced by the two-component expression.

3.2. ISRU pathway

3.2.1. Production schedule and ramp-up

A critical distinction between the two pathways is their delivery timing. Earth manufacturing can begin immediately upon program start, requiring no new infrastructure beyond existing terrestrial supply chains. The ISRU pathway, by contrast, requires construction and commissioning of an extraterrestrial facility before production can begin, followed by a logistic ramp-up to full capacity.

Earth delivery schedule.. The Earth pathway delivers units at a constant rate from $t = 0$:

$$t_{n,E} = \frac{n}{\dot{n}_{\max}}, \quad (7)$$

where $\dot{n}_{\max}$ is the full-capacity production rate (units/year). This assumes pre-existing Earth manufacturing capacity, reasonable given the maturity of terrestrial aerospace production; we test a logistic Earth ramp-up ($t_{0,E} \in \{1, 2\}$ yr) in §Appendix A.

ISRU delivery schedule.. The ISRU facility's instantaneous production rate follows a logistic ramp-up function:

$$\dot{n}(t) = \dot{n}_{\max} \cdot S(t), \quad S(t) = \frac{1}{1 + e^{-k(t-t_0)}}, \quad (8)$$

where t_0 is the ramp-up midpoint (years from program start) and $k = 2.0$ (fixed) controls steepness. Integrating the production rate yields the cumulative production function:

$$N(t) = \frac{\dot{n}_{\max}}{k} [\ln(1 + e^{k(t-t_0)}) - \ln 2], \quad (9)$$

The constant $-\ln 2$ ensures $N(t_0) = 0$. A piecewise formulation enforces $\dot{n}(t) = 0$ for $t < t_c$ ($t_c = t_0 - 1$ by default); at $t = t_0$ the rate is half-maximal, exceeding 90% capacity within ~ 1 year (Figure 7). The closed-form inverse gives the calendar time of the n th ISRU unit:

$$t_{n,I} = t_0 + \frac{1}{k} \ln(2 e^{nk/\dot{n}_{\max, \text{eff}}} - 1). \quad (10)$$

(obtained by inverting Eq. 9). Note that $t_{n,I}$ includes the pre-production construction period: t_0 encompasses both facility construction and commissioning (Eqs. 8–9).

We introduce a transport duration τ_{trans} (baseline 0.5 yr) representing low-energy lunar-to-GEO transfer, so delivery time is

$$t_{n,I}^{\text{del}} = t_{n,I} + \tau_{\text{trans}}. \quad (11)$$

Discounting and revenue-delay calculations use the delivery time $t_{n,I}^{\text{del}}$ rather than the production time $t_{n,I}$. In the Monte Carlo, $\tau_{\text{trans}} \sim U[0.25, 2.0]$ yr, bracketing chemical (~ 0.25 yr) and solar-electric ($\sim 1-2$ yr) low-energy transfer options.

The effective ISRU production rate accounts for facility availability:

$$\dot{n}_{\text{max,eff}} = A \cdot \dot{n}_{\text{max}}, \quad (12)$$

where $A \in [0, 1]$ is the fraction of calendar time the facility is operational. The deterministic baseline uses $A = 1.0$; the Monte Carlo samples $A \sim U[0.70, 0.95]$.

Timing gap.. The separation between these schedules is substantial: when Earth has delivered its 1,000th unit ($t = 2$ yr), ISRU has not yet produced any (its 1,000th unit arrives at $t \approx 7.3$ yr; Table B.27 in Appendix Appendix B). This gap has a counterintuitive NPV consequence: because Earth costs are incurred earlier, they are discounted *less*, making the Earth pathway more expensive in NPV terms. This partially offsets the ISRU pathway's heavy upfront capital burden.

3.2.2. Cost model

For illustrative comparison of per-unit costs, the ISRU Average Unit Cost (AUC) of the n th unit can be expressed as an amortized capital component plus an operational component:

$$C_{\text{ISRU}}^{\text{AUC}}(n) = \frac{K}{N_{\text{total}}} + C_{\text{ops}}(n), \quad (13)$$

where K is the total capital investment required to establish the ISRU infrastructure, and N_{total} is a planning horizon over which capital is amortized for display purposes. We set $N_{\text{total}} = 10,000$ throughout; this choice affects the per-unit cost visualization (Figure 3) but does not enter the crossover calculation, which uses the cumulative formulation below.

The operational cost follows a Wright learning curve with a cost floor C_{floor} representing irreducible per-unit expenses (energy, consumables, remote operations overhead), scaled by a mass penalty factor α and augmented by

a transport cost for delivering ISRU-manufactured units to the operational orbit:

$$C_{\text{ops}}(n) = \alpha [C_{\text{floor}} + (C_{\text{ops}}^{(1)} - C_{\text{floor}}) \cdot n^{b_I}] + m \cdot p_{\text{transport}} \cdot \alpha, \quad (14)$$

where $b_I = \ln(\text{LR}_I)/\ln 2$ with ISRU learning rate LR_I , C_{floor} represents the asymptotic minimum operational cost per unit (energy, consumables, remote operations overhead), $\alpha \geq 1$ is a delivered-mass penalty: an ISRU-manufactured unit with $\alpha = 1.3$ has 30% more mass than the reference design (thicker structural margins, lower-purity alloys) and therefore requires proportionally more processing and transport—both operational cost and transport cost scale with α because both are proportional to the mass actually processed and delivered—and $p_{\text{transport}}$ (\$/kg) is the cost of transporting an ISRU-manufactured unit from the production site to the operational orbit. The cost floor is sampled as $C_{\text{floor}} \sim U[\$0.3\text{M}, \$2.0\text{M}]$ in the Monte Carlo; when C_{floor} exceeds the Earth pathway’s per-unit launch cost floor ($m \cdot p_{\text{launch}} = \1.85M at baseline), the per-unit cost advantage of ISRU is reduced, potentially preventing crossover within the planning horizon. This scenario is captured in the non-convergence statistics (§4.3).

A production yield parameter $Y \in (0, 1]$ captures the fraction of ISRU-manufactured units that pass quality acceptance: delivering one functional unit requires producing $1/Y$ units, so the effective operational cost is $C_{\text{ops}}(n)/Y$. The baseline MC uses $Y = 1.0$ (perfect yield); Y is tested as a deterministic sensitivity parameter (Table 1), not a stochastic MC input, because the effect is small relative to K and LR_E variance ($<0.5\%$ of total output variance at $Y \geq 0.80$). Even at $Y = 0.80$, the crossover shifts by only +259 units (+6.9%).

Table 1: ISRU production yield sensitivity (deterministic, $r = 5\%$, phased K). Baseline $N^* = 3,749$ at $Y = 1.0$.

Y	N^*	Shift	% Shift
1.00	3,749	—	—
0.95	3,801	+52	+1.4%
0.90	3,860	+111	+3.0%
0.85	3,928	+179	+4.8%
0.80	4,008	+259	+6.9%
0.70	4,218	+469	+12.5%

The ramp-up effect is captured purely through timing: early units are produced later and therefore discounted more heavily in the NPV calculation (Eq. 10). The cumulative cost of producing N units via the ISRU pathway is

$$\Sigma_{\text{ISRU}}(N) = K + \sum_{n=1}^N C_{\text{ops}}(n). \quad (15)$$

In the lump-sum formulation, the full capital cost K is incurred at $t = 0$; in the phased baseline (Eq. 23), K is spread over five annual tranches to reflect realistic deployment schedules. The amortized per-unit formulation (Eq. 13) is used only for visualization.

3.2.3. Crossover and NPV formulation

The undiscounted crossover N_0^* is the smallest N with $\Sigma_{\text{ISRU}}(N) \leq \Sigma_{\text{Earth}}(N)$. Because the ISRU pathway concentrates expenditure early while Earth distributes costs over time, each pathway's costs must be discounted according to its own delivery schedule. We define the NPV crossover N^* as the smallest N satisfying

$$\underbrace{\sum_{y=0}^4 \frac{K/5}{(1+r)^{t_0-5+y}}}_{K_{\text{eff}}} + \sum_{n=1}^N \frac{C_{\text{ops}}(n)}{(1+r)^{t_{n,I}^{\text{del}}}} \leq \sum_{n=1}^N \frac{C_{\text{Earth}}(n)}{(1+r)^{t_{n,E}}}, \quad (16)$$

where r is the real discount rate (all costs in constant 2024 USD), $t_{n,I}^{\text{del}} = t_{n,I} + \tau_{\text{trans}}$ is the ISRU delivery time including transport (the code discounts at delivery time; this notation makes the convention explicit), and the five capex tranches are paid during years $[t_0 - 5, t_0]$, ending at commissioning. This coupling ensures that when t_0 is large, capital spending is also deferred. At $r = 5\%$ and $t_0 = 5$, $K_{\text{eff}} = \$45.3\text{B}$ (-9.4% vs. lump-sum K). Because Earth costs are incurred earlier ($t_{n,E} < t_{n,I}^{\text{del}}$), they carry higher present value, making the Earth pathway more expensive in NPV terms; this partially offsets the ISRU pathway's upfront capital burden. When $r = 0$, Eq. 16 reduces to the undiscounted comparison.

Permanent vs. transient crossover classification. A crossover is *permanent* if the ISRU pathway remains cheaper at all subsequent production volumes, and *transient* if the Earth pathway eventually re-crosses at very large n . Formally, the crossover is permanent iff

$$\lim_{n \rightarrow \infty} C_{\text{ISRU}}^{\text{ops}}(n) < \lim_{n \rightarrow \infty} C_{\text{Earth}}(n). \quad (17)$$

The ISRU asymptotic per-unit cost (learning terms $\rightarrow 0$) is

$$C_{\text{ISRU}}^{\infty} = (1 - f_v) [\alpha C_{\text{floor}} + m p_{\text{transport}} \alpha] + f_v m (p_{\text{fuel}} + c_{\text{vit}}),$$

and the Earth asymptotic per-unit cost is

$$C_{\text{Earth}}^{\infty} = C_{\text{mat}} + m p_{\text{fuel}},$$

where the learnable components (C_{labor} , p_{ops}) vanish at large n . When $f_v > 0$, the vitamin component contributes a non-vanishing term ($f_v \cdot m \cdot c_{\text{vit}} \approx \0.93M at baseline), which can raise C_{ISRU}^{∞} above $C_{\text{Earth}}^{\infty}$, making most crossovers transient. The Monte Carlo classifies each run using Eq. 17; see §4.3 for results.

For transient crossovers, we define the *re-crossing volume* N^{**} as the smallest $N > N^*$ at which the ISRU cumulative NPV cost again exceeds the Earth pathway:

$$N^{**} = \min \left\{ N > N^* : \sum_{n=1}^N \frac{C_{\text{ops}}(n)}{(1+r)^{t_{n,I}^{\text{del}}}} + K > \sum_{n=1}^N \frac{C_{\text{Earth}}(n)}{(1+r)^{t_{n,E}}} \right\}. \quad (18)$$

N^{**} is searched up to $N = 200,000$; runs that do not re-cross within this bound are right-censored.² The savings window $[N^*, N^{**}]$ represents the production range over which ISRU is cost-advantaged. We distinguish three categories based on this framework:

- **Asymptotically permanent** (Eq. 17): $C_{\text{ISRU}}^{\infty} < C_{\text{Earth}}^{\infty}$; ISRU per-unit cost is lower at $n \rightarrow \infty$.
- **Finite-horizon permanent**: $N^{**} > 200,000$; does not re-cross within the search bound despite $C_{\text{ISRU}}^{\infty} \geq C_{\text{Earth}}^{\infty}$.
- **Finite-horizon transient**: $N^{**} \leq 200,000$; re-crosses within the search bound.

²With NPV discounting, late-arriving costs are heavily attenuated, so many asymptotically transient scenarios do not re-cross within any practical horizon.

3.2.4. Earth-sourced component fraction

Even passive structural modules require Earth-sourced fasteners, seals, sensors, and coatings. We model this via a vitamin fraction f_v : a fraction f_v of each unit’s mass must be launched from Earth, with its own manufacturing cost c_{vit} (\$/kg). The ISRU operational cost becomes

$$C_{\text{ops}}^{\text{vit}}(n) = (1 - f_v) \cdot C_{\text{ops}}(n) + f_v \cdot m \cdot (p_{\text{launch,eff}}(n) + c_{\text{vit}}), \quad (19)$$

where $p_{\text{launch,eff}}(n)$ includes launch learning when active (i.e., the Earth pathway’s two-component model, Eq. 5), and $c_{\text{vit}} = \$10,000/\text{kg}$ reflects the manufacturing cost of mechanical vitamin components—fasteners, seals, coatings, and simple sensors—rather than radiation-hardened electronics (which can exceed $\$100,000/\text{kg}$ [36]). Cost derivation: aerospace-grade titanium fasteners cost $\$200\text{--}\$800/\text{kg}$ raw plus $\$2,000\text{--}\$5,000/\text{kg}$ for machining, inspection, and certification at lot sizes of $10^2\text{--}10^3$ [36]; thermal coatings and adhesives add $\$1,000\text{--}\$3,000/\text{kg}$; simple sensors (thermocouples, strain gauges) cost $\$3,000\text{--}\$8,000/\text{kg}$ at aerospace volumes. A weighted average across the vitamin BOM (Table D.29) yields $\sim\$8,000\text{--}\$12,000/\text{kg}$, supporting the $\$10,000/\text{kg}$ baseline. If the vitamin fraction includes avionics or rad-hard electronics, c_{vit} would exceed $\$50,000/\text{kg}$, which is tested as a sensitivity case and represents a crossover-preventing failure mode. The $\$10,000/\text{kg}$ figure is thus a scenario parameter representing a “mechanical vitamins only” architecture; the sensitivity analysis tests $c_{\text{vit}} = \$50,000/\text{kg}$ to bound the case where rad-hard electronics are included in the vitamin fraction. The baseline uses $f_v = 0.05$ (5% of unit mass Earth-sourced). This vitamin fraction is included in the baseline because it drives the permanent/transient crossover distinction (§4.3): at large n , the vitamin component’s asymptotic cost ($f_v \cdot m \cdot (p_{\text{fuel}} + c_{\text{vit}}) \approx \0.94M) raises the ISRU floor above the Earth pathway’s asymptotic cost, making most crossovers transient (finite-horizon amortization effects). This is a conservative modeling choice; if vitamin components can eventually be ISRU-sourced, all crossovers become permanent. The $f_v = 0$ case (fully ISRU-derived, an optimistic bound) is reported as a sensitivity variant. We test $f_v \in \{0, 0.05, 0.10, 0.15, 0.20\}$ and $c_{\text{vit}} \in \{\$5,000, \$10,000, \$50,000\}/\text{kg}$ in §4.2. A deterministic sweep of f_v from 0 to 0.20 (at 1% increments) reveals a permanence cliff: the transition from permanent to transient crossover occurs at $f_v \approx 4\%$, corresponding to the point where the vitamin component raises C_{ISRU}^∞ above C_{Earth}^∞ . Below this threshold, all crossovers are permanent; above it, crossovers remain achievable but are finite-horizon amortization effects.

Dynamic vitamin fraction.. In practice, f_v is expected to decline as ISRU capabilities mature: components initially Earth-sourced may become locally producible as additive manufacturing, coating, and sensor fabrication processes are demonstrated in situ. We model this via an exponential decay:

$$f_v(n) = f_v^{\text{floor}} + (f_v^{(0)} - f_v^{\text{floor}}) e^{-n/n_v}, \quad (20)$$

where $f_v^{(0)} = 0.05$ is the initial fraction, $f_v^{\text{floor}} \sim U[0.01, 0.03]$ is the irreducible minimum (mechanical fasteners, seals), and $n_v \sim U[2,000, 10,000]$ is the maturation scale. Both are sampled stochastically in the canonical MC. The dynamic model converts most analytically transient crossovers into functionally permanent ones: at $n = 10,000$, f_v has decayed to within 10% of its floor for $n_v \leq 3,000$, and the savings window extends beyond any practical program horizon. At asymptotically large n , $f_v \rightarrow f_v^{\text{floor}}$ and the permanence condition (Eq. 17) is evaluated using the floor value. In the canonical MC (Table 11), 22% of converged runs are analytically permanent; the remaining 78% are technically transient but have savings window widths exceeding 190,000 units (right-censored at the search horizon), making them functionally indistinguishable from permanent crossovers.

A representative mass breakdown (Table D.29 in Appendix Appendix D) connects the 15% total vitamin content to the 5% irreducible Earth-sourced fraction.

Physical archetype definition.. The baseline unit is an unpressurized structural truss segment: an aluminum-alloy frame with thermal coatings and mechanical fasteners, representative of solar collector support structures or orbital habitat framing. The vitamin fraction $f_v = 0.05$ corresponds to irreducible Earth-sourced fasteners and fittings (titanium bolts, seals, simple sensors). To test sensitivity to the physical archetype, Table 2 reports crossover results for three representative structural product classes spanning the range of complexity and vitamin content likely in large-scale space manufacturing programs.

The unpressurized truss (lowest complexity) achieves permanent crossover at the lowest N^* because its minimal vitamin fraction keeps the ISRU asymptotic floor below Earth's. The habitat module shell (highest complexity) requires the most units but still achieves crossover, confirming that ISRU is economically viable across the structural product spectrum.

Table 2: Physical archetype sensitivity. Each archetype represents a distinct structural product class with characteristic vitamin fraction and vitamin component cost. Crossover computed with phased K (5 yr) at $r = 5\%$.

Archetype	f_v	c_{vit} (\$/kg)	N^*	Permanent?
Unpressurized truss	0.03	5,000	3,564	Yes
Pressurized struct. panel	0.08	10,000	3,923	No
Habitat module shell	0.15	20,000	6,272	No

3.3. Monte Carlo framework

To characterize the sensitivity of N^* to parameter uncertainty, we embedded the NPV cost model in a Monte Carlo simulation. Each of 10,000 runs samples eighteen independent parameter values (plus two derived) from the distributions specified in Table 3 and computes the NPV crossover point via Eq. 16. We define “crossover achievement within horizon H ” as the event $N^* \leq H$; throughout this paper, $H = 40,000$ unless otherwise stated. Tables abbreviate this as “Conv.” for brevity. Critically, the discount rate r is *not* treated as a stochastic parameter. Instead, the full Monte Carlo ensemble is run independently at each of several fixed discount rates ($r \in \{3\%, 5\%, 8\%\}$). This separation is motivated by the observation that the discount rate reflects the decision-maker’s time preference and financing structure rather than technological uncertainty; conflating it with stochastic cost parameters obscures the distinct roles of economic policy and engineering risk [3]. We report the median, interquartile range, and 10th/90th percentiles of the resulting crossover distribution at each rate, with 95% bootstrap confidence intervals computed from 5,000 bootstrap resamples.

The model configuration (Table D.30 in Appendix Appendix D) maps active equations to the baseline MC and sensitivity variants.

Learning rate parameters are sampled from clipped normal distributions (Table 3). Three parameters—launch cost, ISRU capital, and production rate—are correlated via a 3D Gaussian copula ($\rho_{p,K} = 0.3$, $\rho_{K,\dot{n}} = 0.5$, $\rho_{p,\dot{n}} = 0$). The launch–capital correlation captures common factors (regulatory environment, technology maturity). The capital–production rate correlation ($\rho_{K,\dot{n}} = 0.5$) reflects that higher-capacity facilities require larger investment, eliminating implausible corners (e.g., \$100B capital with 250 units/yr); independence ($\rho_{K,\dot{n}} = 0$) is tested as a sensitivity variant (Appendix Appendix A). All other parameters are sampled independently. The sampled

Table 3: Monte Carlo parameter distributions. Launch cost, ISRU capital, and production rate are correlated via a 3D Gaussian copula ($\rho_{p,K} = 0.3$, $\rho_{K,\dot{n}} = 0.5$); all other parameters are sampled independently. ISRU capital follows a log-normal distribution; results are presented at dual baselines $\sigma_{\text{ln}} = 0.70$ (Flyvbjerg terrestrial, P90/P50 $\approx 2.5\times$) and $\sigma_{\text{ln}} = 1.0$ (space-specific, P90/P50 $\approx 4.1\times$). The Monte Carlo is run at fixed discount rates $r \in \{3\%, 5\%, 8\%\}$. Achieving crossover is defined as $N^* \leq H$ where $H = 40,000$. Baseline values are rounded for exposition; the simulation uses exact values as specified in the code repository.

Parameter	Baseline	Distribution	Range
<i>Stochastic parameters (18 independent + 2 derived = 20 total)[¶]</i>			
Launch cost p_{launch} (\$/kg)	1,000	Uniform [†]	[500, 2,000]
ISRU capital K (\$B)	50 [#]	Log-normal [†]	med. 65, $\sigma_{\text{ln}}=0.70$, clip [20, 200]
Earth learning rate LR_E	0.85	$\mathcal{N}(0.85, 0.03)$	[0.75, 0.95]
ISRU learning rate LR_I	0.90	$\mathcal{N}(0.90, 0.03)$	[0.80, 0.98]
Ramp-up time t_0 (years)	5	Uniform	[3, 8]
First-unit ops. cost $C_{\text{ops}}^{(1)}$ (\$M)	5	Uniform	[2, 10]
First-unit mfg. cost $C_{\text{mfg}}^{(1)}$ (\$M)	75	Uniform	[50, 100]
Material cost C_{mat} (\$M)	1.0	Uniform	[0.5, 2.0]
Labor cost $C_{\text{labor}}^{(1)}$ (\$M)	74 [§]	Derived	$C_{\text{mfg}}^{(1)} - C_{\text{mat}}$
Mass penalty α	1.0	Uniform	[1.0, 2.0]
Transport cost $p_{\text{transport}}$ (\$/kg)	100	Uniform	[50, 300]
Ops cost floor C_{floor} (\$M)	0.5	Uniform	[0.3, 2.0]
Production rate $\dot{n}_{\text{max}}$ (units/yr)	500	Uniform	[250, 750]
ISRU availability A	1.0 [‡]	Uniform	[0.70, 0.95]
Launch cost floor p_{fuel} (\$/kg)	200	Uniform	[100, 400]
Learnable launch comp. p_{ops} (\$/kg)	800	Derived	$\max(p_{\text{launch}} - p_{\text{fuel}}, 0)$
Transport duration τ_{trans} (yr)	0.5	Uniform	[0.25, 2.0]
<i>Plateau & vitamin parameters (new in AK)</i>			
Earth plateau onset n_{break}	—	Uniform	[200, 1,000]
Plateau damping η	—	Uniform	[0.3, 0.7]
Vitamin decay scale n_v	—	Uniform	[2,000, 10,000]
Vitamin floor f_v^{floor}	—	Uniform	[0.01, 0.03]
<i>Fixed parameters</i>			
Discount rate r	0.05	Fixed per run	{0.03, 0.05, 0.08}
Unit mass m (kg)	1,850	Fixed	—
S-curve steepness k	2.0	Fixed	—
Amortization horizon N_{total}	10,000	Fixed	—
Launch-capital correlation $\rho_{p,K}$	0.3	Fixed	—
Capital-production rate corr. $\rho_{K,\dot{n}}$	0.5	Fixed	—

[†]Launch cost and capital correlated ($\rho_{p,K} = 0.3$); capital and prod. rate correlated ($\rho_{K,\dot{n}} = 0.5$) via 3D Gaussian copula.

[‡]Deterministic baseline value; the MC samples $A \sim U[0.70, 0.95]$.

[§] $C_{\text{labor}}^{(1)}$ follows the Wright learning curve; C_{mat} is non-learnable (Eq. 2).

^{||} p_{ops} is derived from p_{launch} and p_{fuel} ; it is not independently sampled.

[¶]18 independent + 2 derived stochastic parameters: $C_{\text{labor}}^{(1)}$ is derived ($C_{\text{mfg}}^{(1)} - C_{\text{mat}}$) and p_{ops} is derived ($\max(p_{\text{launch}} - p_{\text{fuel}}, 0)$); both are sampled stochastically via their parent distributions.

Enforces $C_{\text{mfg}}^{(1)} \geq C_{\text{mat}}$ and $C_{\text{ops}}^{(1)} \geq C_{\text{floor}}$; clamping rate <1%.

[#]Deterministic baseline uses \$50B; MC samples from log-normal with median \$65B (Appendix Appendix D).

p_{launch} is decomposed per Eq. 5; the propellant floor p_{fuel} is now sampled stochastically ($U[\$100, \$400]/\text{kg}$) to reflect uncertainty in the GEO delivery cost asymptote. For global sensitivity analysis, we compute Spearman rank correlations and partial rank correlation coefficients between each input parameter and N^* .

3.4. Parameter justification

The most consequential parameter values are summarized here; detailed derivations, empirical analogies, and a rough K subsystem decomposition (~ 500 t of delivered infrastructure at $\sim \$5,000/\text{kg}$ plus 20% integration margin) are in Appendix Appendix D.

ISRU capital K (median \$65B) is sampled log-normal ($\sigma_{\text{ln}} = 0.70$, clipped [$\20B, $\$200\text{B}$]); results are presented at dual baselines $\sigma_{\text{ln}} = 0.70$ (Flyvbjerg terrestrial, $P90/P50 \approx 2.5\times$) and $\sigma_{\text{ln}} = 1.0$ (space-specific, $P90/P50 \approx 4.1\times$). *Learning rates:* $\text{LR}_E \sim \mathcal{N}(0.85, 0.03)$ is calibrated from empirical aerospace data (Table D.31); $\text{LR}_I \sim \mathcal{N}(0.90, 0.03)$ relies on terrestrial additive manufacturing analogy—no direct extraterrestrial data exists.

Empirical grounding assessment. Table 4 summarizes the empirical basis and sensitivity rank for each key parameter. The asymmetry between pathways is substantial: Earth-pathway parameters have moderate-to-strong empirical grounding, while ISRU-pathway parameters rely almost entirely on analogy. Results should be interpreted in light of this asymmetry.

3.5. Assumptions and limitations

Key assumptions are listed here; detailed discussion is in Appendix Appendix E.

- *Scope:* passive structural modules only; system-level crossover depends on structural cost fraction ($\sim 1/f_c$ scaling).
- *Cash-flow timing:* pay-at-milestone for both pathways; manufacturing lead-time sensitivity (-3.6% to -6.6%) confirms conservatism (Appendix Appendix A).
- *Quality parity:* Earth and ISRU units meet identical specs; optimistic for early ISRU production. A yield parameter Y (Table 1) tests the impact of imperfect ISRU quality acceptance.
- *Fixed unit mass:* no design evolution over the production run.

Table 4: Parameter confidence assessment. Grounding: S=strong (direct empirical), M=moderate (cross-domain analogy), W=weak (order-of-magnitude estimate), N=none (pure assumption). Sensitivity from PRCC (Table 15).

Parameter	Pathway	Grounding	Sensitivity
LR_E	Earth	M (sat. production, $n \leq 500$)	Very high
K	ISRU	W (megaproject analogy)	Very high
$C_{\text{mfg}}^{(1)}$	Earth	M (Iridium NEXT, SMAD)	Moderate
$\dot{n}_{\text{max}}$	Shared	W (facility scaling)	Moderate
α	ISRU	N (no empirical basis)	Moderate
LR_I	ISRU	N (no extraterrestrial data)	Moderate
p_{launch}	Earth	S (Falcon 9/Starship data)	Moderate
$C_{\text{ops}}^{(1)}$	ISRU	W (energy + ops estimate)	Moderate
C_{floor}	ISRU	W (energy cost estimate)	Low
p_{fuel}	Earth	M (bottom-up decomposition)	Low

- *Single product type*: multi-product facilities would shift crossover earlier.
- *Constant real discount rate*: risk-adjusted or pathway-specific rates tested as sensitivities (§Appendix A).
- *Earth instant start*: tested via ramp-up robustness checks (§Appendix A).

4. Results

4.1. Baseline crossover

Under baseline parameter values (Table 3, including $f_v = 0.05$), the undiscounted cumulative cost curves ($r = 0$) intersect at approximately $N^* = 3,733$ units. Figure 1 shows these trajectories. The ISRU pathway begins with a steep initial outlay (the \$50B capital investment plus operational and transport costs during ramp-up), while the Earth pathway accumulates cost at a rate dominated by the constant launch cost per unit. The Earth curve is approximately linear at large N because the learning-curve reduction in manufacturing cost is small relative to the constant launch cost per unit (\$1.85M/unit at \$1,000/kg for a 1,850 kg unit). The ISRU curve, by contrast, flattens as operational learning drives per-unit costs toward the cost floor plus the transport cost.

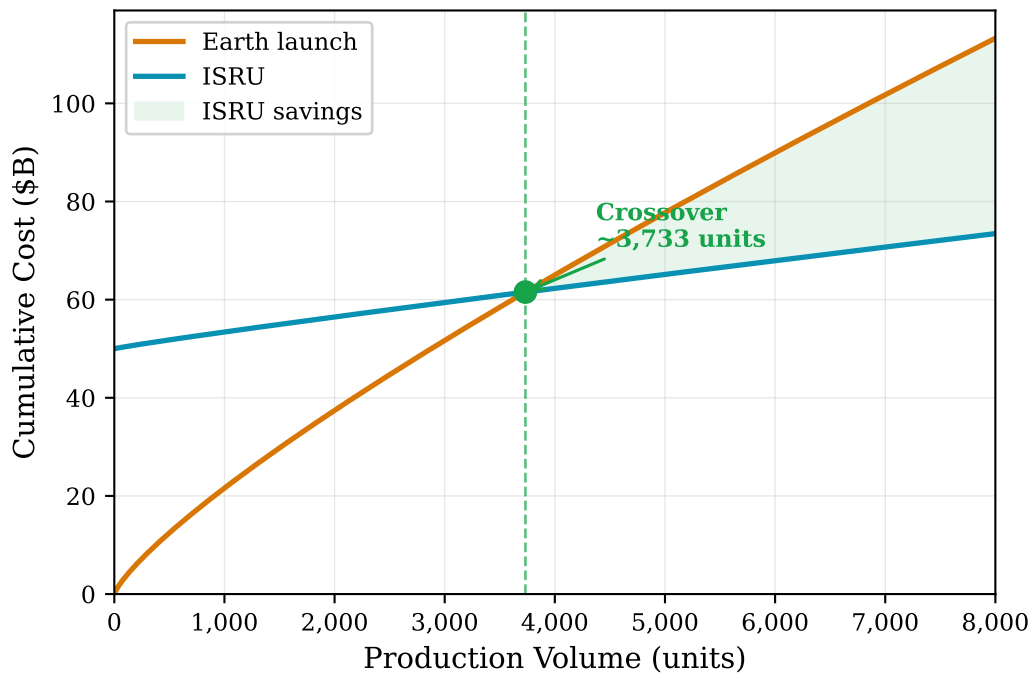


Figure 1: Undiscounted cumulative cost (\$B) vs. production volume for the Earth-launch and ISRU pathways under baseline assumptions ($r = 0$, $f_v = 0.05$). The crossover occurs at approximately 3,733 units. The shaded region indicates cumulative ISRU savings.

Applying NPV discounting at 5% with phased capital deployment (5-year baseline, Eq. 23) and pathway-specific delivery schedules (Eq. 16) yields a crossover of approximately $N^* = 3,749$ units (compared to $N^* = 4,374$ under lump-sum K). Because Earth costs are incurred earlier and thus discounted less, they carry higher present value (see §3), partially offsetting the ISRU capital burden. Figure 2 shows cumulative cost curves at four discount rates. At $r = 10\%$, the crossover reaches $\sim 5,802$ units.

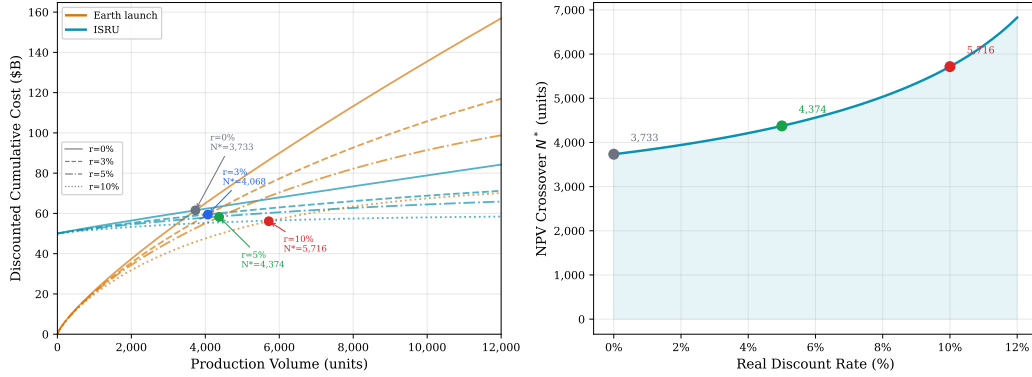


Figure 2: Left: Discounted cumulative cost curves at four discount rates with pathway-specific delivery schedules, with crossover points marked. Right: NPV crossover point as a continuous function of discount rate. The pathway-specific formulation yields lower crossover points than earlier analyses because Earth costs carry greater present-value weight.

Table 5 summarizes the crossover point under representative scenarios at both $r = 0$ and $r = 5\%$.

Table 5: Crossover points under representative scenarios at two discount rates, using pathway-specific delivery schedules and phased capital (5-year baseline).

Scenario	\$/kg	K (\$B)	Undiscounted		NPV ($r = 5\%$)	
			N^*	Time	N^*	Time
Optimistic	500	30	$\sim 2,032$	~ 9 yr	$\sim 1,855$	~ 9 yr
Baseline	1,000	50	$\sim 3,733$	~ 12 yr	$\sim 3,749$	~ 12 yr
Conservative	2,000	100	$\sim 8,029$	~ 21 yr	$\sim 11,608$	~ 28 yr

Product archetype sensitivity. The baseline models a spacecraft-class structural module (\$75M first-unit, 1,850 kg). An industrial structural product

(Archetype B: \$15M first-unit, 3,000 kg, $LR_E = 0.90$, $\alpha = 1.3$) has a higher deterministic crossover but similar MC conditional median ($\sim 5,100$ – $5,300$), confirming that the crossover is not an artifact of the specific product class. Full comparison in Appendix Appendix A.

Figure 3 illustrates the per-unit cost dynamics. The Earth per-unit cost is bounded below by $m \cdot p_{\text{launch}}$, while ISRU declines toward $C_{\text{floor}} = \$0.5\text{M}$ —this asymptotic divergence is the structural driver of the crossover. However, as the Monte Carlo analysis shows (§4.3), crossover is achieved in 75–92% of stochastic scenarios, not in all cases; see §4.3 for the permanent/transient crossover breakdown. An Earth pathway sanity check against satellite production economics is provided in Appendix Appendix D.

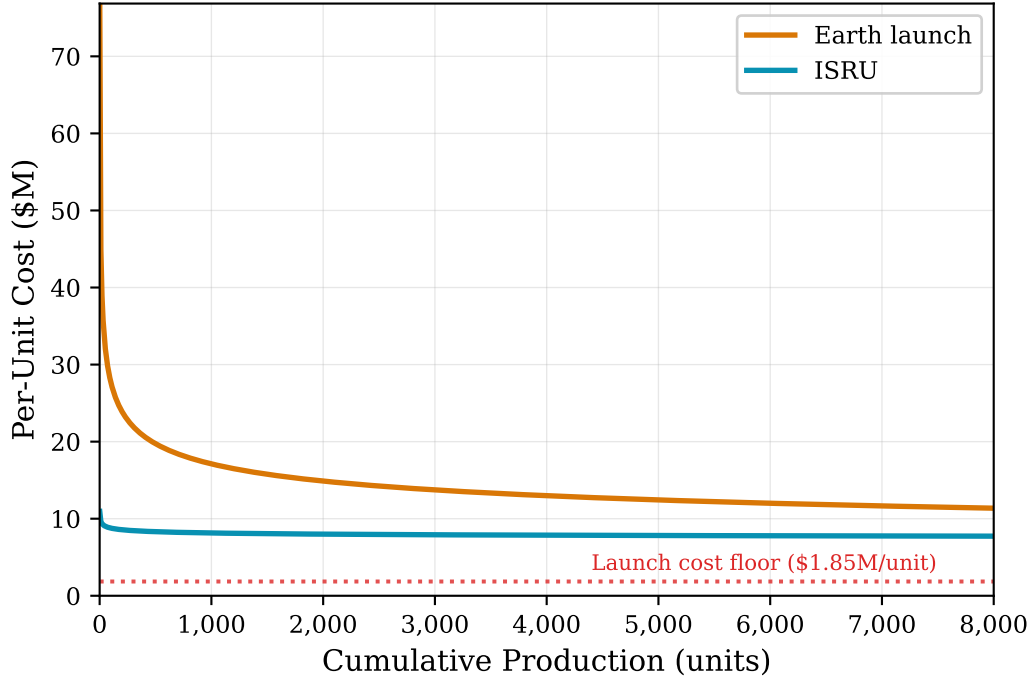


Figure 3: Per-unit cost (\$M) vs. cumulative production for the Earth-launch and ISRU pathways. The Earth pathway is bounded below by the launch cost floor ($m \cdot p_{\text{launch}} = \1.85M), while the ISRU pathway declines toward the operational cost floor ($C_{\text{floor}} = \$0.5\text{M}$).

4.2. Sensitivity analysis

Figure 4 summarizes one-at-a-time sensitivity of the NPV crossover ($r = 5\%$) to nine parameters.

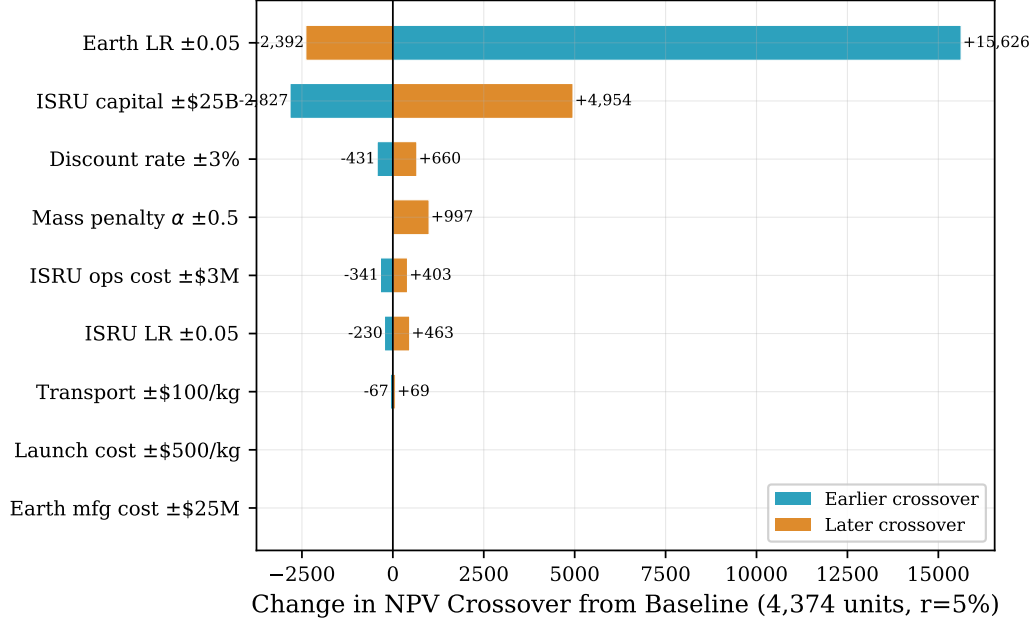


Figure 4: Tornado diagram showing the ranked sensitivity of the NPV crossover point ($r = 5\%$ baseline, pathway-specific timing) to each input parameter.

The Earth manufacturing learning rate remains the most influential parameter in absolute range, with a strongly asymmetric effect: a *higher* learning rate ($\text{LR}_E = 0.90$, i.e., slower learning—each doubling reduces cost by only 10%) shifts the crossover earlier because the Earth pathway retains higher costs for longer, while a *lower* rate ($\text{LR}_E = 0.80$, faster learning) delays crossover because Earth manufacturing costs decline rapidly. ISRU capital investment is the second most influential parameter. The discount rate r remains important but its effect is moderated under pathway-specific timing because the timing gap between pathways partially offsets the discounting penalty.

The mass penalty factor α (varied from 1.0 to 2.0) shifts the crossover by approximately +440 units at $\alpha = 1.5$. Transport cost $p_{\text{transport}}$ has a smaller effect, consistent with transport being a modest fraction of total ISRU cost at baseline parameters. First-unit costs ($C_{\text{ops}}^{(1)}$, $C_{\text{mfg}}^{(1)}$) exert moderate influence.

ISRU learning rate and launch cost remain relatively weak individual drivers, though their effects compound with other parameters in the Monte Carlo analysis.

The two-dimensional parameter space of launch cost versus ISRU capital is shown in the heat map of Figure 5, now computed using the NPV crossover at $r = 5\%$ with pathway-specific timing.

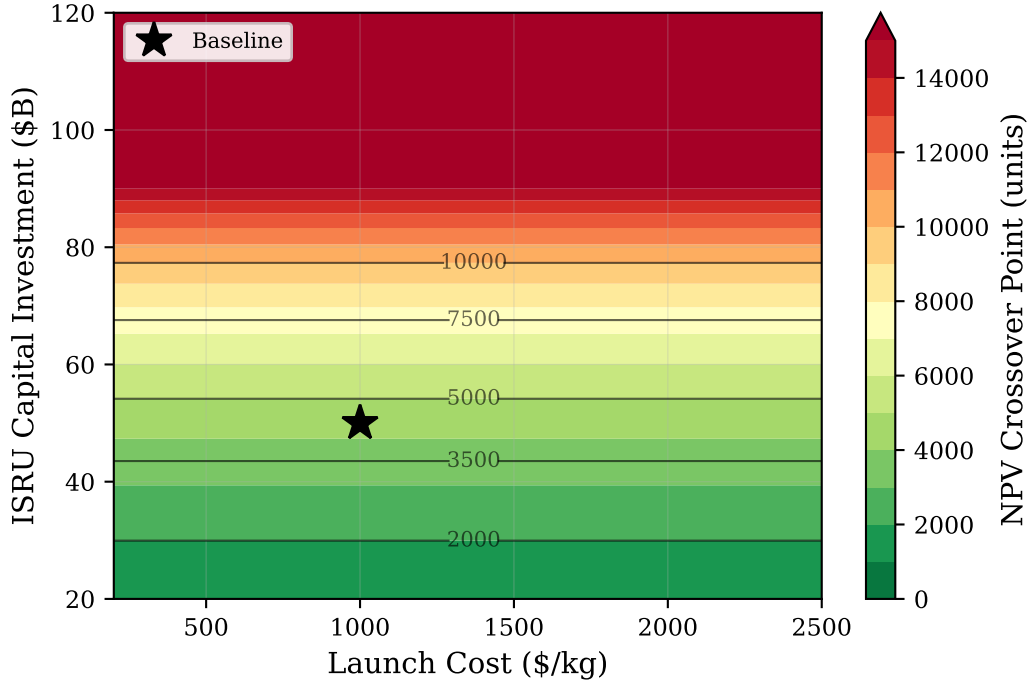


Figure 5: NPV crossover point N^* ($r = 5\%$, pathway-specific timing) as a function of launch cost (\$/kg) and ISRU capital investment (\$B). The star marks the baseline scenario.

Launch cost learning sweep.. The baseline deterministic case uses constant launch cost (Eq. 4); the baseline MC uses the two-component model (Eq. 5, Table D.30). As a sensitivity variant, we test the two-component learning model (Eq. 5) with LR_L ranging from 0.90 (aggressive: 10% cost reduction per doubling) to 0.97 (moderate). Table A.23 reports the results. Even at the aggressive $LR_L = 0.90$ —where the ops component falls to $\sim \$334/\text{kg}$ by $n = 10,000$ —the crossover shifts by only +315 units from baseline (+7%). The no-learning case ($LR_L = 1.00$) gives N^* identical to the respective baseline because launch learning has negligible effect at baseline scale. The reason

launch learning cannot eliminate the ISRU advantage is structural: even at $LR_L = 0.90$, the fuel component (\$200/kg) constitutes an assumed operational asymptote under Earth-supplied propellant logistics. At $n = 10,000$, the ops component has fallen to $\sim \$334/\text{kg}$, but the combined launch cost ($\$534/\text{kg} \times 1,850 \text{ kg} = \0.99M per unit) remains above the ISRU operational cost floor. This floor asymmetry—the assumed operational asymptote under high-cadence reusable launch versus experience-driven ISRU operational costs—is the fundamental reason that launch learning, regardless of its rate, cannot fully close the gap at high production volumes. The p_{fuel} floor is architecture-dependent, not physics-fundamental: tug learning could reduce the GEO delivery component, and next-generation propellant technologies could lower the propellant cost itself. The $U[\$100, \$400]/\text{kg}$ range brackets this uncertainty; a tug-learning scenario (§4.2) tests the effect of aggressive operations learning ($LR_L = 0.90$) and reduced propellant floors ($\$50\text{--}100/\text{kg}$). The ISRU propellant scenario below extends this analysis to the case where ISRU-produced propellant reduces the fuel floor itself.

Launch learning sweep results (Table A.23 in Appendix Appendix A) confirm that even at $LR_L = 0.90$, the crossover shifts by only +315 units (+7%).

ISRU learning rate boundary scenarios. To bound the sensitivity of the crossover to the ISRU learning rate—an assumption with limited empirical grounding (see §3.4)—we test two extreme scenarios. With *no ISRU learning* ($LR_I = 1.0$, operational costs remain constant at $C_{\text{ops}}^{(1)}$), the NPV crossover shifts to $\sim 5,962$ units (+1,588 from baseline). With *very slow learning* ($LR_I = 0.98$), the crossover shifts to $\sim 5,370$ units (+996). In both cases the crossover still occurs, demonstrating that the ISRU economic advantage does not depend on optimistic learning assumptions—it is driven primarily by the amortization of fixed capital against a declining per-unit cost, even in the absence of experience-curve effects.

Learning curve plateau model. Empirical evidence suggests that Wright curve learning rates moderate at high cumulative volumes [2, 34, 5], with regime transitions typically at $n \sim 100\text{--}500$ in aerospace. Since the crossover occurs at $n \sim 3,700$ —well beyond this empirical base—we integrate a piecewise plateau model *stochastically* in the canonical MC: $n_{\text{break}} \sim U[200, 1,000]$ and $\eta \sim U[0.3, 0.7]$ are sampled independently for each run. The plateau replaces the standard unit labor cost $C_{\text{labor}}(n) = C_{\text{labor}}^{(1)} \cdot n^{b_E}$ with a continuous

piecewise function:

$$C_{\text{labor}}(n) = \begin{cases} C_{\text{labor}}^{(1)} \cdot n^{b_E} & n \leq n_{\text{break}} \\ C_{\text{labor}}^{(1)} \cdot n_{\text{break}}^{b_E(1-\eta)} \cdot n^{b_E\eta} & n > n_{\text{break}} \end{cases} \quad (21)$$

where $\eta \in (0, 1]$ is the damping factor (reduced learning exponent) and n_{break} is the transition volume. Both branches evaluate to $C_{\text{labor}}^{(1)} \cdot n_{\text{break}}^{b_E}$ at $n = n_{\text{break}}$, ensuring cost continuity. The pure Wright curve ($\eta = 1$) serves as an optimistic bound; the distinction between intra-program learning (one production line, $n \leq 500$ empirical base) and inter-industry learning (PV modules to >100 GW [13], wind turbines [28]) is critical.

The stochastic plateau *strengthens* the ISRU case: by moderating Earth’s learning beyond the empirical regime, it raises Earth costs relative to the pure Wright extrapolation, increasing the crossover achievement rate from 74% (pure Wright) to 85%. The plateau parameters contribute modestly to overall variance ($R^2 < 0.5\%$ each; Table 15) but are significant conditional drivers of crossover location (n_{break} : PRCC = +0.28; η : PRCC = +0.20). Deterministic sweep results ($n_{\text{break}} \in \{200, 500, 1000, 2000\}$, $\eta \in \{0.3, 0.5, 0.7\}$) are in Appendix Appendix A.

K tail sensitivity (σ_{ln} variation).. The baseline uses $\sigma_{\text{ln}} = 0.70$ for the log-normal K distribution (P90/P50 $\approx 2.5\times$), at the conservative end of Flyvbjerg’s 2–4 $\times$ range [12]. To test whether heavier tails materially affect conclusions, we vary $\sigma_{\text{ln}} \in \{0.70, 1.0, 1.3\}$ (Table 6).

Table 6: K tail sensitivity: effect of log-normal spread on crossover achievement (10,000 runs, $r = 5\%$, $H = 40,000$). *Variant configuration*: σ_{ln} varied from canonical baseline (0.70). The [\$20B, \$200B] clip bounds truncate the heaviest tails at $\sigma_{\text{ln}} \geq 1.0$.

σ_{ln}	Conv. %	Cond. median	P90/P50(K)	K P10	K P90
0.70	85.1%	4,311	2.50 $\times$	\$26.9B	\$163.4B
1.00	80.8%	3,754	3.05 $\times$	\$20.0B	\$200.0B
1.30	77.8%	3,258	3.05 $\times$	\$20.0B	\$200.0B

Table 7 presents the dual-baseline headline results at both σ_{ln} values across discount rates.

Heavier tails reduce the convergence rate modestly (85% $\rightarrow$ 78% from $\sigma_{\text{ln}} = 0.70$ to 1.3) as more extreme K draws exceed the planning horizon.

Table 7: Dual-baseline headline results: crossover statistics at $\sigma_{\text{ln}} = 0.70$ (terrestrial reference class) and $\sigma_{\text{ln}} = 1.0$ (space-specific reference class), 10,000 MC runs, $H = 40,000$. *Note:* this table uses a σ_{ln} -variation ensemble; the canonical baseline results are in Table 11.

σ_{ln}	Label	Conv. %	Cond. median	Sav. win. @20k
0.70	Terrestrial	85.1%	4,311	94.7%
1.00	Space-specific	80.8%	3,753	93.7%

The conditional median *decreases* because the wider left tail also includes more low- K scenarios that converge at smaller N^* . At $\sigma_{\text{ln}} \geq 1.0$, the [\$20B, \$200B] clip bounds become binding (Table 6), truncating the theoretical tails. Crucially, the savings window probability is nearly invariant to σ_{ln} : 94.7% vs. 93.7% at $N_h = 20,000$ (Table 7), confirming that the planning-relevant metric is robust to distributional assumptions.

Earth-sourced component fraction.. At $c_{\text{vit}} = \$10,000/\text{kg}$, the crossover is preserved up to $f_v = 20\%$ (+1,492 units); at $c_{\text{vit}} = \$50,000/\text{kg}$, it is eliminated. The crossover requires only that ISRU produce the structural bulk mass; full component substitution is not necessary.

Propellant floor and launch evolution.. The baseline propellant floor ($p_{\text{fuel}} = \$200/\text{kg}$) is an assumed operational asymptote, not a physics-fundamental limit. Under aggressive tug learning ($\text{LR}_L = 0.90$) with reduced propellant floor (\$50/kg), the crossover shifts by only +332 units, because the ISRU advantage arises from capital amortization, not from the launch cost floor level.

Secondary sensitivity tests.. Over 30 additional robustness tests—including pioneering phase costs ($\gamma = 2\text{--}5\times$), QA costs, K -clip bounds, Earth scaling penalties, manufacturing cost floors, production heritage (n_0), and Iridium NEXT cross-check—are reported in Appendix Appendix A. Among these secondary tests, no single one shifts the crossover by more than 7%; the top-driver tests (LR scenarios, plateau, vitamins) produce larger shifts up to $\pm 38\%$ but are reported individually above. Table 8 indexes all tests.

4.3. Monte Carlo robustness

Table 9 defines the canonical baseline configuration that produces all headline statistics reported in the abstract and conclusion. All tables and

Table 8: Index of sensitivity tests. “Main” tests retained in §4.2–§5; “App.” tests in Appendix Appendix A. The five top drivers and three failure modes are in the main text.

Test	Location	Max shift	Category
<i>Main text (top drivers + failure modes)</i>			
Learning rate scenarios (LR_E , LR_I)	§4.2	+1,588	Top driver
Learning plateau (Earth, stochastic)	§4.2	−1,424	Top driver
Vitamin sensitivity (f_v , c_{vit})	§4.2	+1,492	Top driver
Revenue breakeven (R^*)	§5	—	Decision metric
Technology disruption	§5.5	+3,583	Disruption
Product archetype	§4.2	—	Robustness
Earth scaling penalty (γ_E)	§4.2	−720 (−16.5%)	New (AD)
Failure: $c_{vit} > \$50k/kg$	§4.2	>40k	Failure mode
Failure: $r > 20\%$	§4.7	>40k	Failure mode
Failure: $p_s < \text{threshold}$	§4.6	—	Failure mode
Asymmetric learning plateau	§4.2	+167	Robustness
Hybrid transition strategy	§5.2.1	—	Decision metric
Physical archetype (f_v , c_{vit})	§3.2.4	varies	New (AG)
Tug learning scenario	§4.2	+332	New (AG)
Triangular prior sensitivity	App. Appendix A	−446 med.	New (AG)
K -median sweep	§4.3	—	Primary (AL)
Logistic learning saturation	§4.3	+491	Model-form (AL)
ISRU production yield (Y)	§3	+469 ($Y=0.70$)	New (AL)
Savings window bootstrap CI	§4.5	± 1.2 pp	Precision
<i>Appendix (secondary tests)</i>			
Copula ρ sensitivity	App. Appendix A	<200	Minor
K – $\dot{n}$ correlation	App. Appendix A	−185	Minor
Piecewise schedule	App. Appendix A	0	Negligible
Cash-flow timing	App. Appendix A	−6.6%	Minor
Earth capex (K_E)	App. Appendix A	−1,187	Favors ISRU
C_{floor} failure threshold	App. Appendix A	—	Analytic
Additional correlations	App. Appendix A	<200	Minor
No-launch-learning	App. Appendix A	± 0	Negligible
Fuel floor decomposition	App. Appendix A	± 44	Negligible
Launches-per-unit	App. Appendix A	± 15	Negligible
Maintenance (ϕ_K)	App. Appendix A	+5,515	Moderate
Capex coupling	App. Appendix A	<100	Minor
K -ramp (phased capital)	App. Appendix A	−630	Minor
Earth mfg floor ($C_{\text{mfg}}^{\text{floor}}$)	App. Appendix A	0	Negligible

figures in this paper use this configuration unless explicitly noted otherwise.

Table 9: Canonical baseline Monte Carlo configuration. All headline statistics (abstract, conclusion, Tables 11–12, 19) correspond to this configuration. Tables using variant configurations are labeled explicitly.

Setting	Value
K distribution	Log-normal, median \$65B, $\sigma_{\ln} = 0.70$
K clip bounds	[\$20B, \$200B]
Capex phasing	5 annual tranches, coupled to t_0
Learning model	Wright with stochastic plateau (n_{break}, η)
Vitamin fraction	Dynamic: $f_v(n) = f_v^{\text{floor}} + (0.05 - f_v^{\text{floor}})e^{-n/n_v}$; $c_{\text{vit}} = \$10,000/\text{kg}$
Discount rate	Fixed real r (not stochastic); headline at $r = 5\%$
Planning horizon	$H = 40,000$ units
Copula	3D Gaussian ($\rho_{p,K} = 0.3$, $\rho_{K,\dot{n}} = 0.5$)
Conditioning	Conditional statistics report converging runs only
Other parameters	Table 3

Table 10 maps the principal configurations used in this paper to their respective crossover values, resolving any ambiguity between deterministic and Monte Carlo baselines.

Table 10: Configuration-to-crossover mapping. “Det.” = deterministic baseline ($K = \$50\text{B}$); “MC” = canonical Monte Carlo (Table 9, K median \$65B). Phased = 5-year coupled capex; lump = K at $t = 0$.

Configuration	K treatment	Det. N^*	MC cond. median
Undiscounted, lump	\$50B at $t = 0$	3,733	—
NPV $r = 5\%$, lump	\$50B at $t = 0$	4,374	—
NPV $r = 5\%$, phased coupled	\$50B phased/ t_0	3,749	—
Canonical MC, $r = 5\%$	LN med. \$65B, phased	—	4,311

The full Monte Carlo simulation (10,000 runs per rate) samples all eighteen independent stochastic parameters (plus two derived)—including four new plateau and dynamic vitamin parameters—simultaneously from the distributions in Table 3, with the copula correlations specified in Table 9, and computes the NPV crossover at each of three fixed discount rates. Table 11 summarizes the results across rates.

First, the probability of achieving crossover within H decreases monotonically with r : 92% at $r = 3\%$ versus 75% at $r = 8\%$, reflecting the increasing

Table 11: Monte Carlo results at fixed discount rates (10,000 runs each, canonical baseline per Table 9). “Conv.” = fraction of scenarios achieving crossover within $H = 40,000$.

r	Conv. %	Cond. median	Cond. IQR	Cond. P10	Cond. P90
3%	91.9%	4,772	[2,501, 9,365]	1,434	17,020
5%	85.1%	4,311	[2,256, 8,370]	1,308	14,860
8%	75.0%	3,596	[1,891, 6,848]	1,129	12,024

NPV penalty on the ISRU pathway’s upfront capital. The stochastic plateau contributes roughly 11 percentage points to the crossover achievement rate (85% vs. 74% at $r = 5\%$ under pure Wright), because moderating Earth’s learning makes the ISRU pathway relatively cheaper. The wider log-normal K distribution reduces convergence rates by ~ 8 percentage points relative to the uniform distribution (Appendix Appendix A). Second, the conditional median is stable across rates ($\sim 3,600$ – $4,800$), indicating that among scenarios that do achieve crossover, the crossover point is relatively insensitive to the discount rate. The discount rate primarily affects *whether* crossover is achieved rather than *where* it occurs—high rates push marginal scenarios past the planning horizon.

Table 12 presents the empirical cumulative distribution function $P(N^* \leq H)$ at selected production horizons, providing a survival-analysis-style characterization of the crossover distribution.

Table 12: Probability of achieving crossover at or below production horizon H , $P(N^* \leq H)$.

Horizon H (units)	$r = 3\%$	$r = 5\%$	$r = 8\%$
1,000	3.9%	4.6%	5.8%
2,000	16.8%	18.1%	20.2%
5,000	47.7%	47.7%	47.6%
10,000	70.7%	68.6%	64.5%
20,000	85.5%	80.6%	72.8%

Bootstrap confidence intervals (5,000 resamples) on the conditional median at $r = 5\%$ yield a 95% CI of [4,198, 4,423]. The distribution (Figure 6) is heavily right-skewed at all rates.

Epistemic vs. parametric uncertainty. The Monte Carlo propagates *parametric* uncertainty—variation within the assumed model structure and prior distributions—and includes two model-form extensions: stochastic learning saturation (n_{break}, η) and dynamic vitamin maturation (f_v^{floor}, n_v). Further model-form uncertainty remains: alternative cost structures (quality-adjusted costs, design evolution) could shift the probability surface. The crossover occurs at $n \sim 3,700\text{--}4,400$, an order of magnitude beyond the empirical aerospace base ($n \leq 500$); while the stochastic plateau directly addresses this, the functional form is a maintained assumption. To assess model-form sensitivity, Table 13 compares the piecewise plateau against a logistic saturation alternative:

$$C_{\text{labor}}(n) = C_{\text{mat}} + \frac{C_{\text{Wright}}(n) - C_{\text{mat}}}{1 + (n/n_{\text{half}})^2}, \quad (22)$$

where $C_{\text{mat}} = m \cdot p_{\text{fuel}}$ is the irreducible material cost (the natural labor-cost floor for Earth manufacturing), n_{half} is the half-saturation volume, and the quadratic exponent produces a sharper saturation transition than a linear Hill function. The logistic form has been tested at three deterministic parameter values; full stochastic integration would require specifying prior distributions for n_{half} and is deferred to future work. Headline probabilities should be interpreted as “given these priors and this model structure, $X\%$ of parameter space achieves crossover,” not as frequentist confidence in crossover per se.

Table 13: Learning curve model-form sensitivity (deterministic, $r = 5\%$, phased K). “Plateau” = piecewise damped exponent; “Logistic” = smooth saturation at n_{half} . Pure Wright (no saturation) yields $N^* = 3,749$.

Model	$n_{\text{break}}/n_{\text{half}}$	Damping	N^*	Shift
Plateau	200	0.3	2,286	−1,463
Plateau	500	0.5	2,716	−1,033
Plateau	1,000	0.7	2,939	−810
Logistic	200	—	3,843	+94
Logistic	500	—	4,240	+491
Logistic	1,000	—	4,218	+469

The logistic form produces crossover points within $\pm 13\%$ of the pure Wright baseline, while the piecewise plateau shifts crossover earlier by 22–39%. The qualitative conclusion is robust: both forms yield crossover in

the $\sim 2,300$ – $4,300$ range. However, the forms differ in direction: the plateau strengthens the ISRU case while the logistic form slightly weakens it. This model-form uncertainty ($\sim \pm 1,500$ units) is comparable to the K uncertainty and should be considered alongside the parametric results.

Variance decomposition.. First-order rank-regression R^2 values confirm that the model is effectively low-dimensional: K alone explains 62.6% of output variance (unconditional) and LR_E an additional 20.0%, yielding a cumulative $R^2 = 0.82$ from just two parameters. Adding production rate and $C_{\text{mfg}}^{(1)}$ brings the cumulative to 91%; the full 18-parameter regression (excluding 2 derived parameters) achieves $R^2 = 0.94$. Conditional on convergence, K ($R^2 = 0.59$) and production rate ($R^2 = 0.14$) dominate, with LR_E at $R^2 = 0.11$ and total model $R^2 = 0.94$. The four new plateau and vitamin parameters collectively explain $< 0.5\%$ of variance individually, confirming that their primary effect is on the *probability* of crossover achievement (+11 pp) rather than on its *location*; the stochastic p_{fuel} explains $< 0.1\%$. These results validate the PRCC rankings (Table 15) and confirm that reducing uncertainty in K and LR_E would have the highest value-of-information payoff.

K-conditional crossover surface.. Because K explains 63% of variance yet rests on order-of-magnitude estimates, the headline probabilities are best understood as conditional on the assumed K prior. Table 14 presents crossover economics as a continuous function of K median, enabling readers to apply their own capital assumptions.

Table 14: K -median sweep (10,000 MC runs, $r = 5\%$, $H = 40,000$, $\sigma_{\text{ln}} = 0.70$). “Det. N^* ” uses lump-sum K at the listed median value (not phased); compare Table 10 for phased baselines. The $K = \$65\text{B}$ row matches the canonical baseline. If $K < \$75\text{B}$, ISRU crossover is highly probable ($> 80\%$); if $K > \$100\text{B}$, it drops below 67%.

K median	Det. N^* (lump)	Conv. %	Cond. median	Cond. IQR
\$50B	4,374	92.1%	3,268	[1,687, 6,461]
\$65B	6,952	85.1%	4,311	[2,256, 8,370]
\$75B	9,328	80.0%	4,928	[2,612, 9,492]
\$100B	22,993	66.8%	6,306	[3,376, 11,438]
\$150B	$> 40\text{k}$	45.5%	8,387	[4,755, 14,608]

MC convergence diagnostic.. The conditional median stabilizes within $\pm 2\%$ of its final value by 5,000 runs, confirming that 10,000 runs provide adequate statistical power.

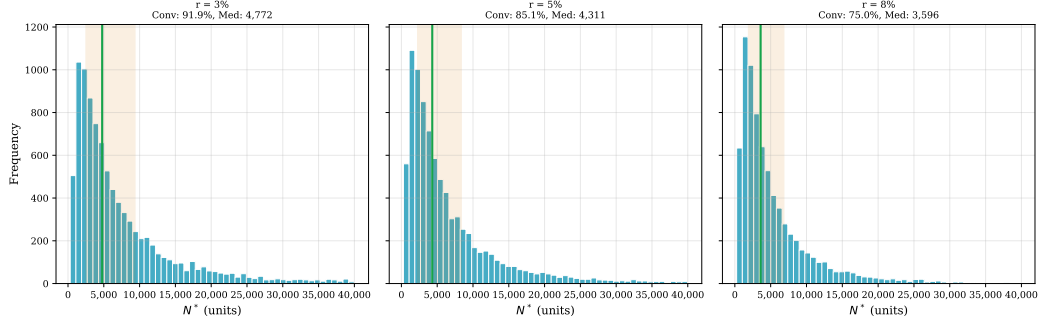


Figure 6: Distribution of NPV crossover points (N^*) across 10,000 Monte Carlo runs at three fixed discount rates. Each panel shows scenarios achieving crossover within $H = 40,000$; the title reports the fraction achieving crossover and the conditional median. The vertical line marks the conditional median; the shaded band shows the conditional IQR.

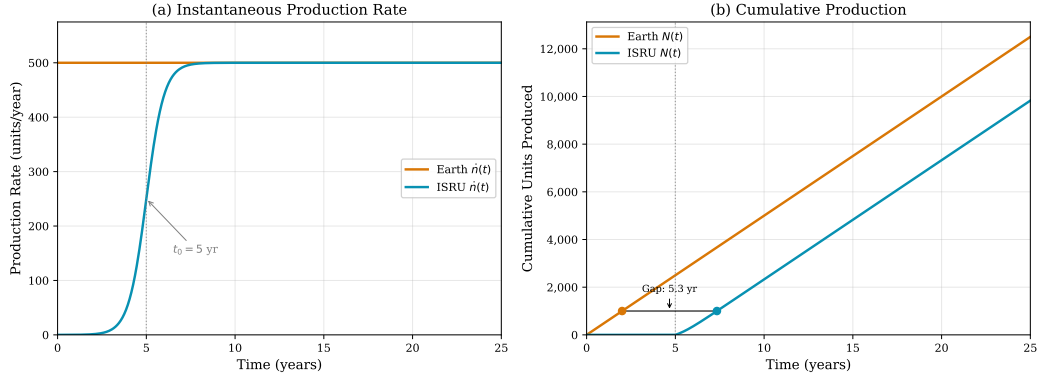


Figure 7: Production schedule verification. (a) Instantaneous production rate for both pathways: Earth produces at full rate from $t = 0$, while ISRU follows a logistic ramp-up with $t_0 = 5$ yr. (b) Cumulative production, showing the timing gap between pathways (5.3 yr at $n = 1,000$). The ISRU facility reaches half-maximal rate at t_0 and 90% capacity by $t_0 + 1.1$ yr.

Table 15 presents Spearman rank correlations between each input parameter and N^* at $r = 5\%$, computed both unconditionally and conditional on achieving crossover.

Table 15: Sensitivity rankings at $r = 5\%$: Spearman rank correlations (ρ_S) and partial rank correlation coefficients (PRCC). PRCC is the primary metric as it controls for inter-parameter correlations, resolving confounding from the copula. “Cond.” includes only converging scenarios. All $|\rho_S| > 0.03$ and $|\text{PRCC}| > 0.03$ are significant at $p < 0.001$.

Parameter	ρ_S (cond.)	PRCC (cond.)	PRCC (unc.)	Interpretation
K	+0.52	+0.95	+0.95	Higher capital delays
LR_E	−0.54	−0.92	−0.91	Slower Earth learning → earlier
t_0	−0.02	−0.50	−0.47	Later ramp-up delays
$C_{\text{ops}}^{(1)}$	+0.07	+0.35	+0.33	Higher ISRU ops delays
$\dot{n}_{\text{max}}$	+0.31	−0.32	−0.38	Higher rate → earlier [†]
n_{break}	+0.05	+0.28	+0.26	Later plateau → delays
α	+0.04	+0.28	+0.26	Mass penalty delays
LR_I	+0.03	+0.27	+0.24	Slower ISRU learning delays
p_{launch}	+0.21	−0.24	−0.23	Higher launch → earlier
η	+0.05	+0.20	+0.21	Stronger damping delays
C_{floor}	+0.03	+0.14	+0.14	Higher floor delays
$p_{\text{transport}}$	+0.01	+0.07	+0.06	Minor
A	+0.00	+0.04	+0.07	Minor

[†]Sign reversal ($\rho_S > 0$ vs. $\text{PRCC} < 0$) reflects confounding with K via the copula ($\rho_{K,\dot{n}} = 0.5$); partialing out K reveals the direct effect: higher production rate → earlier crossover.

The PRCC column (controlling for inter-parameter correlations) provides the preferred sensitivity ranking. K and LR_E are the dominant drivers, consistent with the variance decomposition (§4.3). The 14.9% of non-achieving scenarios at $r = 5\%$ are strongly associated with high ISRU capital; the wider log-normal tail produces more extreme K draws that prevent convergence. The stochastic plateau parameters (n_{break} , η) rank 6th and 10th by conditional PRCC, confirming their role as secondary but significant drivers of crossover location.

Two-part sensitivity decomposition.. Since 15% of runs at $r = 5\%$ never achieve crossover, the PRCC conflates two questions: what drives *whether* crossover is achieved (binary), and *where* it occurs (continuous). We separate these via point-biserial correlations (Part A) and conditional PRCCs (Part B); K and LR_E remain dominant in both parts, confirming that the conditional ranking is not a selection artifact.

Permanent versus transient crossovers.. When the asymptotic ISRU per-unit cost exceeds the asymptotic Earth per-unit cost, any crossover is a finite-horizon amortization artifact that would reverse at very large production volumes (see the re-crossing analysis in §4.5). With the dynamic vitamin fraction model, the asymptotic vitamin contribution uses the floor $f_v^{\text{floor}} \sim U[0.01, 0.03]$ rather than the initial $f_v = 0.05$, substantially increasing the permanent fraction. At $r = 5\%$: of the 8,511 converging runs, 1,889 (22.2% of converging) achieve permanent crossover, while 6,622 (77.8%) achieve transient crossover. At $r = 3\%$: 2,019 permanent (22.0%), 7,170 transient (78.0%). At $r = 8\%$: 1,658 permanent (22.1%), 5,846 transient (77.9%). For committed programs with finite production horizons, both permanent and transient crossovers represent real economic benefits; the distinction matters primarily for programs contemplating indefinite production scales.

Under positive discount rates, late costs are heavily attenuated; the stochastic plateau and dynamic vitamin fraction together make virtually all transient scenarios functionally permanent. At $r = 5\%$, the median savings window width is $\sim 196,000$ units and 100% of transient runs have $N^{**} > 100,000$ (Table 18). For decision purposes, “transient” denotes only the asymptotic classification (Eq. 17); within any realistic planning horizon, the transient/permanent distinction is moot.

Distribution sensitivity. The baseline uses a log-normal K distribution (median \$65B, $\sigma_{\ln} = 0.70$), calibrated to Flyvbjerg’s megaproject reference class [12]. As a sensitivity variant, replacing the log-normal with a uniform distribution $K \sim U[\$30B, \$100B]$ changes the convergence rate and conditional median modestly, confirming robustness to the distributional form. Replacing all other uniform distributions with triangular (mode at baseline) shifts the conditional median by <300 units. Both shifts are small relative to the IQR.

Copula sensitivity ($\rho_{p,K} \in \{0, 0.3, 0.6\}$) produces <200 units variation in conditional median. An extended 6D copula correlating K with t_0 , A , and $C_{\text{ops}}^{(1)}$ shifts convergence by -0.5 pp (Appendix Appendix A, Table A.24).

Kaplan-Meier survival analysis. Since 15–25% of runs do not achieve crossover within H , conditioning on convergence changes the estimand and can bias comparisons across discount rates (where censoring varies). The Kaplan-Meier (KM) estimator [17] accounts for right-censored non-converging runs, yielding an unbiased median estimate (Table 16).

Table 16: Kaplan-Meier median vs. conditional median (10,000 MC runs, $H = 40,000$, canonical configuration). The KM estimator accounts for right-censored non-converging observations.

r	Conv. %	Cond. median	KM median	Divergence
3%	91.9%	4,772	5,334	11.8%
5%	85.1%	4,311	5,350	24.1%
8%	75.0%	3,596	5,425	50.8%

At $r = 5\%$, the KM median ($\sim 5,350$) exceeds the conditional median ($\sim 4,300$) by 24%, reflecting the upward bias from ignoring censored (non-converging) runs. Divergence increases with censoring: 12% at $r = 3\%$ versus 51% at $r = 8\%$. For *committed programs* (decision already made to pursue a fixed production target), the conditional median is the appropriate planning statistic. For *portfolio-level decisions* (choosing whether to invest in ISRU vs. alternatives), the KM median is more appropriate because it accounts for the possibility that crossover may not occur.

4.4. Cumulative economics and phased capital

Table 17 contextualizes the crossover in absolute terms. The ISRU pathway requires greater expenditure through approximately year 12–13 (the “in-

vestment valley”); by year 20, ISRU saves $\sim \$41\text{B}$ undiscounted ($\$20\text{B}$ NPV at $r = 5\%$). The NPV investment valley extends to year 15, reinforcing that the ISRU decision requires patient capital.

Table 17: Cumulative cost comparison under baseline assumptions, both undiscounted and NPV-discounted ($r = 5\%$). Units produced follow the ISRU S-curve schedule. Production is negligible before year 7 ($t < t_0 + 2$); see Table B.27 for the delivery schedule.

Year	Units	Undiscounted			NPV ($r = 5\%$)		
		Earth (\$B)	ISRU (\$B)	Net (\$B)	Earth (\$B)	ISRU (\$B)	Net (\$B)
10	$\sim 2,327$	42.2	57.4	-15.2	38.3	55.2	-16.9
15	$\sim 4,827$	75.6	64.7	$+10.9$	61.9	59.1	$+2.8$
20	$\sim 7,327$	105.6	71.6	$+33.9$	78.6	62.1	$+16.5$

In practice, ISRU capital would be deployed incrementally; the baseline MC uses phased capital as the primary formulation. K is spread over five annual tranches coupled to the construction schedule (Eq. 16). At the deterministic baseline ($t_0 = 5$), the coupled form reduces to:

$$K_{\text{eff}} = \sum_{y=0}^4 \frac{K/5}{(1+r)^y}. \quad (23)$$

At $r = 5\%$, $K_{\text{eff}} = \$45.3\text{B}$ (-9.4%). More generally, tranches are paid during years $[t_0 - 5, t_0)$, ending at commissioning; this coupling ensures that when t_0 is large, spending is also deferred, avoiding the NPV bias of paying at $t = 0$ for a facility that commissions at $t = t_0$. At $t_0 = 5$, coupled and uncoupled spending are identical; the coupling matters in the MC, where $t_0 \sim U[3, 8]$. The phased-coupled formulation yields a deterministic crossover of $\sim 3,749$ units (compared to $\sim 4,374$ units under lump-sum K).

Ongoing capital maintenance. Periodic maintenance at $\phi_K \cdot K$ every Δt_K years shifts the crossover by $+1,435$ ($\phi_K = 5\%$) to $+5,515$ ($\phi_K = 15\%$); full results in Appendix Appendix A. At $\phi_K = 15\%$, recurring capex deepens the ISRU investment valley by up to 11% of K , elevating the financial commitment required before savings accumulate.

4.5. Cost floor and production rate sensitivity

ISRU cost floor threshold. A deterministic sweep of C_{floor} confirms crossover for all values up to \$3.0M (the stochastic upper bound). Analytically, permanent crossover requires $C_{\text{floor}} < m \cdot p_{\text{launch}} - m \cdot p_{\text{transport}} \cdot \alpha = \1.67M at baseline; above this, crossovers are finite-horizon amortization effects (see re-crossing analysis below). Full analytical derivation and C_{floor} sweep results are in Appendix Appendix A.

Re-crossing analysis. For programs of finite scale, the relevant statistic is the savings window $[N^*, N^{**}]$, where N^{**} is the production volume at which the ISRU cumulative cost again exceeds the Earth pathway. When C_{floor} exceeds the analytical threshold (\$1.67M at baseline) or the vitamin fraction raises the ISRU asymptotic floor above the Earth per-unit cost, any crossover is a finite-horizon amortization effect—not a long-run equilibrium—and applies only within the production horizon over which ISRU’s lower early-unit costs (relative to Earth’s high first-unit manufacturing costs) outweigh ISRU’s higher asymptotic per-unit costs. Table 18 characterizes the re-crossing behavior across the Monte Carlo ensemble for transient crossover runs.

Table 18: Re-crossing statistics for transient crossover runs ($r = 5\%$, 10,000 MC runs). N^{**} is the production volume at which the Earth pathway becomes cheaper again in cumulative NPV terms.

Metric	Value
Transient runs (count)	6,622 (66.2% of total; 77.8% of converging)
Median N^{**} (units)	200,000 [†]
N^{**} IQR	[200,000, 200,000] [†]
Savings window width median	~196,000
Peak savings (\$B)	74.0

[†]Censored at $N = 200,000$; NPV discounting suppresses late re-crossing.

Table 19 converts the transient classification into a decision-relevant metric: for a program committing to N_h units, what is the probability that N_h falls within the ISRU savings window? Under the stochastic plateau and dynamic vitamin model, all transient crossovers are *functionally permanent*: the median N^{**} is right-censored at 200,000 with a savings window width of ~196,000 units (Table 18). NPV discounting combined with the reduced asymptotic vitamin fraction suppresses re-crossing so effectively that no tran-

sient run re-crosses within 100,000 units. The savings window probability (Table 19) is therefore the appropriate primary planning metric.

Table 19: Savings window survival: probability that a program of N_h units falls within the ISRU savings window $[N^*, N^{**}]$ (all converging MC runs, $r = 5\%$). For permanent crossovers, $N^{**} = 200,000$ (effectively infinite). Bootstrap 95% CIs from 5,000 resamples of converging runs.

Planning horizon N_h	$P(N^* \leq N_h \leq N^{**})$	95% bootstrap CI
5,000	56.1%	[55.0%, 57.1%]
10,000	80.6%	[79.7%, 81.4%]
20,000	94.7%	[94.2%, 95.1%]
50,000	100.0%	[100.0%, 100.0%]
100,000	100.0%	[100.0%, 100.0%]

Production rate sensitivity. At 250 units/yr, the crossover shifts by +2,165 units (extended ISRU ramp-up delay); at 1,000 units/yr, by -523 (faster amortization). The sensitivity is asymmetric: low rates penalize ISRU heavily, while high rates show diminishing returns as the timing gap saturates.

4.6. Technical success probability

The preceding Monte Carlo distributions represent performance *conditional on the ISRU system reaching operational status*; the p_s framework below captures the binary probability of achieving that status, covering risks outside the MC parameter space (e.g., launch failure during facility delivery, fundamental process infeasibility). These two risk layers are complementary: MC stochastic parameters (cost, schedule, learning rates) characterize uncertainty *given* an operational system, while p_s captures the binary question of whether the system reaches operations at all. Partial technical failure—where the facility operates but at reduced throughput or degraded quality—manifests within the MC layer as low availability (A) or high first-unit cost ($C_{\text{ops}}^{(1)}$), which in turn lowers the savings S and raises $p_s^{\min}$. The preceding analysis conditions on the ISRU system working as designed. In practice, ISRU for structural manufacturing at the scale assumed here is TRL 3–5; the probability of successful deployment is substantially less than unity. We model this via a simple expected-value framework: ISRU succeeds with probability p_s , yielding savings S relative to the Earth pathway; with probability $(1 - p_s)$, it fails entirely and the sunk capital K is lost (the program reverts

to Earth-only). The ISRU pathway is preferred when $p_s \cdot S > (1 - p_s) \cdot K$, which gives a minimum success probability:

$$p_s^{\min} = \frac{K}{S + K}. \quad (24)$$

At the baseline ($K = \$50\text{B}$, $S = \$22.7\text{B}$ NPV savings at $2N^*$ units), $p_s^{\min} = 68.8\%$. Critically, this threshold depends on the evaluation horizon. Table 20 shows how $p_s^{\min}$ varies with the production volume used to evaluate savings.

Table 20: Minimum technical success probability $p_s^{\min}$ as a function of evaluation horizon ($r = 5\%$, $K = \$50\text{B}$).

Evaluation horizon	NPV savings	$p_s^{\min}$
5,000 units	\$3.9B	92.7%
$1.5N^*$ ($\sim 6,604$)	\$13.0B	79.4%
$2N^*$ ($\sim 8,806$)	\$22.7B	68.8%
10,000 units	\$27.0B	64.9%
$3N^*$ ($\sim 13,209$)	\$35.9B	58.2%
20,000 units	\$46.8B	51.7%

The threshold ranges from 93% (near the crossover, where savings are small) to 52% (at large horizons, where savings are large). The 69% value at $2N^*$ represents a moderate evaluation horizon; programs planning for longer-term production benefit from a lower success probability threshold. This variation reflects the all-or-nothing failure model: with partial success, salvage value, or parallel Earth production, $p_s^{\min}$ would be lower at all horizons. Below this threshold, the expected cost of ISRU failure exceeds the expected savings. For comparison: the probability of a comparable-scale engineering program succeeding on its first attempt—analogue to TRL 6 to 9 transition for a novel manufacturing system in an extreme environment—is historically in the 30–70% range for first-of-kind space systems [36]. At $p_s = 50\%$, the expected net cost is +\$13.6B (Earth preferred); at $p_s = 80\%$, it is −\$8.2B (ISRU preferred). This analysis underscores that ISRU is not a free option: the capital at risk is substantial, and a realistic assessment of technical risk is essential to any investment decision.

4.7. Commercial discount rate

Government space programs typically evaluate investments at social discount rates ($r = 3\text{--}5\%$). Private or commercial ventures may require substan-

tially higher hurdle rates to attract capital. At $r = 10\%$, the deterministic crossover extends to $N^* \approx 5,802$; at $r = 12\%$, to $N^* \approx 6,976$; at $r = 15\%$, to $N^* \approx 15,903$. At $r = 20\%$, *no crossover is achieved within the 40,000-unit horizon*. This finding has important implications: the ISRU economic case is viable under government, quasi-government, and even moderately commercial financing structures, but does not close under aggressive commercial terms. A 20% hurdle rate imposes such severe time-value penalties on the deferred ISRU operational savings that the upfront capital investment cannot be recovered within a plausible production horizon. Notably, the crossover is achievable at $r = 15\%$ (a typical commercial hurdle rate for infrastructure projects), though only at $\sim 15,900$ units—roughly $3.6\times$ the baseline crossover. This means that commercial ISRU is feasible in principle for large-scale programs but may require either (a) a reduction in capital costs through technology maturation, (b) risk mitigation through phased deployment and technology demonstration, or (c) blended public-private financing structures that reduce the effective discount rate to improve the crossover economics.

5. Discussion

Under the canonical baseline (Table 9), the decision-relevant metric for finite-horizon programs is the savings window probability: for a program committing to 20,000 units, 95% of parameter draws fall within the ISRU savings window at $r = 5\%$ (Table 19). While 85% achieve crossover, the stochastic plateau and dynamic vitamin fraction make virtually all transient crossovers functionally permanent (savings window width $\sim 196,000$ units), so the savings window probability closely tracks the crossover rate. Results are also presented at $\sigma_{\text{in}} = 1.0$ (space-specific reference class) in Table 7. A minimum technical success probability of $\sim 69\%$ is required for ISRU to be preferred in expectation (§4.6), and the crossover does not close under commercial discount rates above $\sim 20\%$ (§4.7).

5.1. Throughput and demand context

At megastructure scales, a constraint beyond economics emerges: Earth-to-orbit throughput. At 500 Starship launches per year ($\sim 50,000$ t/yr to orbit), a 10^5 -unit program would require ~ 3.7 years of dedicated capacity—two orders of magnitude above current global launch mass [20]. ISRU relaxes this constraint because feedstock bypasses the gravity well. Quantitative

throughput cap tests (Appendix Appendix A) confirm the crossover is insensitive to Earth delivery caps at baseline scales, but throughput limitations would reinforce the ISRU case at $>10,000$ units.

Demand context.. The crossover volume ($\sim 3,750$ units under phased capital, or $\sim 6,900$ tonnes) corresponds to a 1–2 GW space solar power installation. Programs at the 10 GW scale ($\sim 27,000$ units) are well past crossover; programs below $\sim 2,000$ units (1 GW demo) would not reach it.

5.2. Optimal transition strategy

The hybrid model (§5.2.1) quantitatively confirms the advantage of a phased strategy: Earth manufacturing produces units until N^* , then ISRU takes over. At $N = 10,000$, the option value is negative (ISRU capital not recovered); it becomes positive at $N \geq 20,000$ (Table 21). The critical implication is that ISRU investment should begin well before the crossover to ensure capacity is available when the economics favor it.

5.2.1. Hybrid transition strategy

The preceding analysis compares single-pathway strategies (pure Earth vs. pure ISRU). A rational planner would instead pursue a *hybrid* strategy: manufacture the first N^* units via the Earth pathway (available immediately), then switch to ISRU for subsequent units. The “option value” of ISRU investment is the NPV cost difference between the Earth-only pathway and the hybrid strategy:

$$V_{\text{option}} = \Sigma_{\text{Earth}}^{\text{NPV}}(N) - \left[\Sigma_{\text{Earth}}^{\text{NPV}}(N^*) + K_{\text{eff}} + \sum_{n=N^*+1}^N \frac{C_{\text{ops}}(n)}{(1+r)^{t_{n,I}^{\text{del}}}} \right]. \quad (25)$$

Table 21 reports the hybrid strategy economics at baseline parameters. At $N = 10,000$, the option value is *negative* ($-\$13.6\text{B}$): the ISRU capital invested for only $\sim 6,250$ post-switch units cannot be recovered. The hybrid becomes value-positive at $N \geq 20,000$, where accumulated ISRU savings exceed K_{eff} . This scale dependence is a key practical finding: ISRU investment is economically justified only for programs committing to production well beyond N^* .

Under the hybrid strategy, the ISRU deployment delay penalty applies only to post-switch units ($n > N^*$). The hybrid R_{hybrid}^* is lower than the pure-ISRU R^* because the hybrid strategy saves less (Earth costs for early units

Table 21: Hybrid transition strategy: Earth for units 1– N^* , ISRU for units $N^* + 1$ – N . Phased K (5 yr), $r = 5\%$. Option value = Earth-only NPV – hybrid NPV.

N_{total}	N_{switch}	Hybrid (\$B)	Earth-only (\$B)	Option value (\$B)	R_{hybrid}^*
10,000	3,749	105.1	91.5	–13.6	—
20,000	3,749	109.8	116.1	+6.3	\$0.22M
50,000	3,749	112.5	128.6	+16.1	\$0.45M

are higher than ISRU ops but lower than ISRU capital + ops per unit at small N). However, the option value remains positive at $N \geq 20,000$, confirming that ISRU investment adds value for programs beyond the crossover horizon.

5.2.2. Revenue breakeven: deployment delay vs. cost savings

This finding fundamentally qualifies the headline crossover result and is particularly consequential for revenue-generating infrastructure such as space solar power. ISRU’s multi-year ramp-up delay means Earth-manufactured units are available years earlier; the opportunity cost of the delay can outweigh cost savings.

If each unit generates revenue R (\$/year) beginning at delivery for a finite asset lifetime L years, the NPV of revenue from the Earth pathway exceeds that from the ISRU pathway by the opportunity cost of the delay. The delay for unit n is $\delta_n = t_{n,I}^{\text{del}} - t_{n,E}$, where $t_{n,I}^{\text{del}}$ includes the transport duration (Eq. 11). The exact lost-revenue NPV for unit n is

$$\Delta R_n = R \cdot \frac{(1+r)^{-t_{n,E}} - (1+r)^{-(t_{n,E} + \min(\delta_n, L))}}{\ln(1+r)}, \quad (26)$$

which integrates the continuous revenue stream over the delay interval, properly discounting each infinitesimal revenue increment (the continuous-discounting formulation is exact for revenue accruing continuously; it converges to the discrete sum for annual payments). Summing over all units and solving for R^* :

$$R^*(L) = \frac{\Sigma_{\text{Earth}}^{\text{NPV}}(N) - \Sigma_{\text{ISRU}}^{\text{NPV}}(N)}{\sum_{n=1}^N [(1+r)^{-t_{n,E}} - (1+r)^{-(t_{n,E} + \min(\delta_n, L))}] / \ln(1+r)}, \quad (27)$$

i.e., the exact breakeven revenue rate is the ratio of NPV cost savings to the total discounted lost-revenue annuity factor. Here R denotes the net revenue per unit per year (revenue minus operating costs); Eq. 27 is exact for

continuous revenue streams and neglects only revenue-dependent learning or scale effects. Figure 8 shows N^* as a function of R for several asset lifetimes.

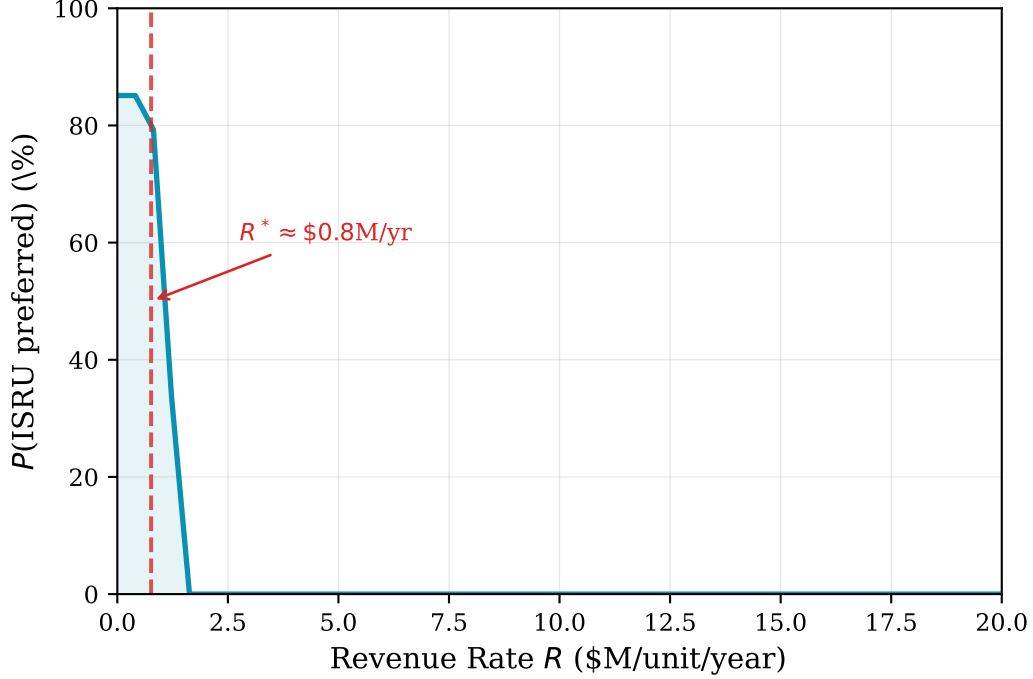


Figure 8: Crossover point N^* as a function of per-unit revenue rate R (\$M/unit/yr) for asset lifetimes $L \in \{10, 20, 30\}$ yr ($r = 5\%$). Above the breakeven revenue R^* , the Earth pathway is preferred on a utility-maximizing basis despite higher cost.

The breakeven revenue $R^* \approx \$0.94\text{M/unit/yr}$ is insensitive to asset lifetime for $L \geq 10$ yr (because the ISRU delay $\bar{\delta} \approx 5.3$ yr is well below these lifetimes); it rises to $\$1.84\text{M/yr}$ only at $L = 3$ yr, irrelevant for structural modules with expected lifetimes > 20 yr. For block-deployed infrastructure (where revenue begins only after a minimum constellation size N_{block} is reached), Eq. 27 overstates the delay penalty because only the last units to complete a block incur opportunity cost; the effective R^* for block deployment exceeds the per-unit value reported here. A quantitative treatment requires specifying the block commissioning schedule and is deferred to future work.

At $\$2\text{M/yr}$ revenue per unit (plausible for space solar power components), the opportunity cost exceeds the ISRU savings ($\$22.7\text{B}$), and the Earth pathway is preferred on a utility-maximizing basis despite being more expensive on a cost-minimizing basis. This finding underscores that the ISRU decision

depends critically on whether the infrastructure is cost-driven (patient capital, no time pressure) or revenue-driven (commercial, time-to-market matters). Under the hybrid strategy (§5.2.1), the delay penalty applies only to post-transition units; the hybrid option value becomes positive at $N \geq 20,000$ (Table 21). The opportunity cost penalty is maximized in a single-program scenario; if the ISRU infrastructure serves as a shared utility across multiple programs, the delay penalty is amortized and R^* shifts upward, expanding the domain in which ISRU is preferred even for revenue-generating infrastructure.

5.3. Implications for near-term space policy

Three policy implications follow. First, *ISRU technology demonstrations are high-value investments*: ISRU investment hedges against stalled Earth manufacturing learning, which would bring the crossover earlier by thousands of units. Second, *launch cost reduction and ISRU investment are complementary*: even at aggressive launch learning ($LR_L = 0.90$), the crossover shifts by only +7%, and cheap launch is essential for deploying ISRU infrastructure. Third, *the discount rate primarily affects crossover achievement probability, not location*: the conditional median is stable across $r \in [3\%, 8\%]$ while achievement probability ranges from 92% to 75%, suggesting public-private partnership as the natural financing structure.

5.4. Decision framework

Figure 9 summarizes the practical decision logic for selecting between the Earth and ISRU pathways, integrating the key branching criteria identified in this analysis: production volume relative to N^* , discount rate regime, technical success probability, revenue generation rate, and vitamin fraction.

5.5. Technology disruption scenarios

Two deterministic disruption scenarios test robustness to technology step-changes at $n = 2,000$: (1) Earth manufacturing cost halved ($N^* = 7,986$, shift +81%); (2) launch cost drops to \$500/kg ($N^* = 4,600$, shift +4.5%). The manufacturing disruption significantly delays crossover but does not prevent it; the launch cost disruption has negligible effect because the structural manufacturing asymmetry, not launch cost, drives the result. When disruptions occur *after* the crossover ($n = 5,000$), both produce zero shift. Full details in Appendix Appendix A.

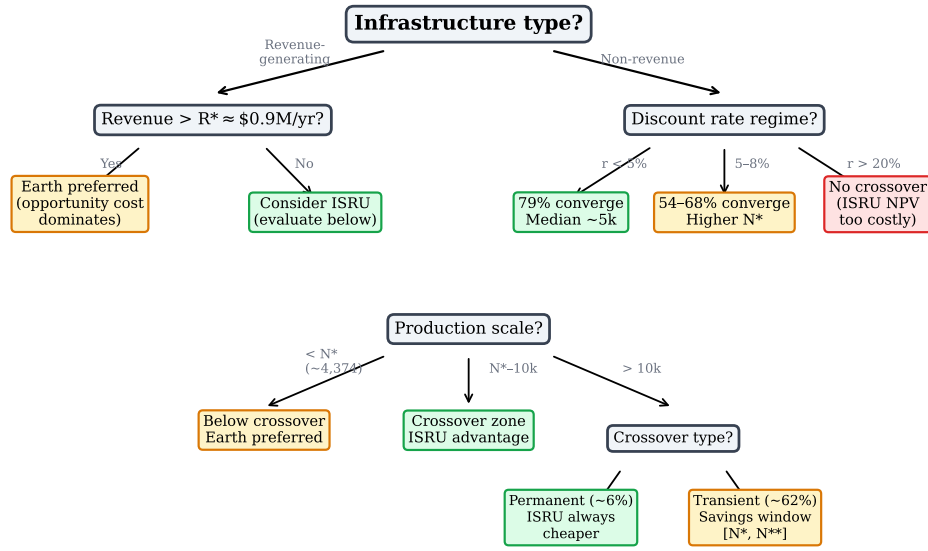


Figure 9: Decision framework for ISRU vs. Earth pathway selection, summarizing the key branching criteria identified in this analysis. Branch thresholds are model-derived: R^* from Eq. 27; discount rate regimes from MC crossover achievement rates (Table 11); N^* from deterministic baseline (Table D.30). Thresholds are illustrative; real decisions require updated data.

5.6. Limitations and future work

The dominant model parameters— K (PRCC = +0.95) and LR_E (PRCC = −0.92)—have been partially grounded: LR_E is cross-checked against Iridium NEXT production data (implied $LR_E \approx 0.85$, §4.2), and the K distribution is calibrated to Flyvbjerg’s megaproject reference class [12]. However, both lack bottom-up engineering calibration for the specific product class modeled; anchoring K to a reference ISRU architecture with subsystem-level cost estimates would further strengthen the analysis.

ISRU pathway empirical grounding gap. The ISRU pathway lacks an empirical anchor comparable to the Earth pathway’s Iridium NEXT cross-check. The ISRU learning rate ($LR_I = 0.90$) relies on terrestrial analogy; three conservative tests mitigate this: no-learning ($LR_I = 1.0$, crossover preserved), pioneering phase ($\gamma = 2\text{--}5\times$), and symmetric learning plateau. A direct cross-check against sub-process ISRU cost models remains an important priority.

The model assumes a fixed unit design over the production horizon (“frozen design”), standard for commodity structural products. For ISRU, this is conservative (design improvements would reduce α and f_v); for Earth, mildly optimistic (disruptive manufacturing advances could reduce $C_{\text{mfg}}^{(1)}$ by a step change). Technology disruption scenarios (§5.5) partially address this; the dynamic vitamin fraction model (§3.2.4) captures supply-chain maturation, but stochastic disruptions with uncertain timing are deferred to future work.

Planned extensions include: real options analysis with stochastic K and t_0 with abandonment optionality; multi-product modeling with shared ISRU infrastructure (shifting crossover earlier); calibration against sub-process ISRU models; and a two-stage decision tree with partial success and salvage value (lowering p_s^{min}).

6. Conclusion

Given the assumed priors and model structure (Table 9), 95% of parameter draws place a 20,000-unit program within the ISRU savings window at $r = 5\%$ (95% CI: [94%, 95%]). Crossover is achieved in 85% of draws (conditional median $\sim 4,300$; KM median $\sim 5,350$; 95% CI: [4,198, 4,423]). Of converging runs, 22% are analytically permanent; the remaining 78% are functionally permanent (savings window width $\sim 196,000$ units). However, these probabilities are dominated by ISRU capital K (63% of variance),

which rests on order-of-magnitude estimates: at K median \$65B, crossover probability is 85%; at \$100B, it drops to 67%; at \$150B, to 46% (Table 14). The headline results are therefore best read as a conditional surface over K , not as unconditional predictions. Model-form sensitivity (Table 13, Eq. 22) confirms qualitative robustness: alternative learning saturation forms shift the crossover by $\pm 1,500$ units without eliminating it; the logistic form and yield parameter (Y) are tested deterministically only. Three factors prevent crossover: vitamin costs $> \sim \$50,000/\text{kg}$, discount rates $> \sim 20\%$, or $p_s \lesssim 70\%$. For revenue-generating infrastructure, deployment delay offsets cost savings above $\sim \$0.94\text{M}/\text{unit}/\text{yr}$ —a finding particularly consequential for space solar power. The ISRU advantage is strongest for non-revenue infrastructure. A hybrid strategy—Earth-first, switching to ISRU at N^* —is value-positive at program scales $\geq 20,000$ units.

Code Availability

The Python simulation code (`isru_model.py`, `isru_mc.py`, `generate_isru_figures.py`) and associated test suite are available at <https://github.com/project-dyson> under an open-source license. Results in this paper were generated from version AM of the codebase (commit PENDING; random seed: 42). A DOI-archived snapshot with a `requirements.txt` and reproduction recipe will be deposited upon acceptance.

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This research was conducted as part of Project Dyson, a non-profit open research initiative. The simulation code is available at <https://github.com/project-dyson>.

Conflicts of Interest

The authors declare no competing interests. This research received no external funding.

Appendix A. Supplementary Sensitivity Tests

Table A.22 summarizes sensitivity tests with individually small or moderate impact on the crossover. Detailed descriptions follow.

Table A.22: Summary of robustness tests (NPV, $r = 5\%$, phased- K baseline $N^* = 3,749$; lump-sum $N^* = 4,374$). Shifts reported relative to lump-sum baseline for comparison with previous versions.

Test	Max shift	Direction
S-curve steepness ($k = 0.5\text{--}4.0$)	± 5	Negligible
Launch learning re-indexing ($\lambda = 0.5\text{--}2.0$)	± 15	Negligible
Rate-dependent learning (20–30% threshold)	0	None
Piecewise construction schedule ($t_c = t_0 - 1$)	0	None
ISRU pre-purchase timing ($\tau = 0.5\text{--}1.0$ yr)	+18	Delays
Earth mfg floor ($C_{\text{mfg}}^{\text{floor}} \leq \3M)	0	None
Fuel floor decomposition (fixed total)	± 44	Negligible
K - $\dot{n}$ correlation ($\rho = 0$ vs. 0.5)	−185	Earlier
Earth-side fixed costs ($K_E = \$1\text{--}10\text{B}$)	−1,187	Earlier (favors ISRU)
Earth scaling penalty ($\gamma_E = 1.1\text{--}1.3$)	−301 to −720	Earlier (favors ISRU)
Extended 6D copula ($K\text{--}t_0$, $K\text{--}A$, $K\text{--}C_{\text{ops}}^{(1)}$)	−0.5 pp	Negligible
Asymmetric plateau (Earth continues, ISRU stalls)	+167	Minor

Earth ramp-up robustness.. Replacing the instant-start Earth schedule with a logistic ramp-up shifts the crossover by +12.3% ($t_{0,E} = 1$ yr) to +23.2% ($t_{0,E} = 2$ yr)—well within the MC range, confirming that the instant-start assumption does not materially bias results.

Risk-adjusted discounting.. Applying an ISRU-specific risk premium ($\Delta r = 2\text{--}5\%$) counterintuitively *reduces* the crossover (−111 to −232 units) because heavier discounting of deferred ISRU operational costs outweighs the capital burden. However, this captures only cash-flow timing risk. The dominant ISRU risks—capital cost overruns, schedule delays, and outright technical failure—operate through mechanisms (higher K , extended t_0 , stranded capital) that would *reverse* this effect.

S-curve steepness.. Sweeping $k \in \{0.5, 1.0, 2.0, 3.0, 4.0\}$ shifts the crossover by ± 5 units; the crossover occurs well past the ramp-up phase.

Launch learning re-indexing.. Testing $\lambda \in \{0.5, 1.0, 2.0\}$ (co-manifest vs. dedicated launches) shifts the crossover by ± 15 units.

Launch cost learning sweep (detail)..

Table A.23: Launch cost learning sweep (NPV, $r = 5\%$, lump-sum K). The fuel component (\$200/kg) is held constant; the operational component (\$800/kg) follows a Wright learning curve.

LR_L	b_L	N^*	Shift	Note
1.00	0.000	4,374	± 0	No learning
0.97	-0.044	4,374	—	Baseline
0.93	-0.105	4,576	+202	
0.90	-0.152	4,689	+315	Aggressive

Fuel floor sensitivity.. Sweeping p_{fuel} from \$50 to \$400/kg with fixed total launch cost shifts the crossover by ± 44 units; the model is insensitive to the fuel-vs-operations split.

Rate-dependent learning.. A rate-dependent modifier freezing the learning curve below 20–30% capacity has zero effect under baseline ramp-up ($k = 2.0$).

Piecewise construction schedule.. Enforcing $\dot{n}(t) = 0$ for $t < t_c = t_0 - 1$ produces no crossover shift; the exponentially small early production is negligible.

Cash-flow timing sensitivity.. Earth manufacturing lead times of 0.5–1.0 yr shift the crossover by -3.6% to -6.6% (earlier), confirming the pay-at-delivery simplification is slightly conservative.

ISRU operational pre-purchase timing.. Shifting ISRU operational costs earlier by 0.5–1.0 yr shifts the crossover by $+0.3\%$ to $+0.4\%$, negligible compared to the Earth timing effect.

Earth-side fixed costs.. Adding Earth-side capital $K_E = \$1\text{--}10\text{B}$ shifts the crossover by -129 to $-1,187$ units, uniformly favoring ISRU.

ISRU cost floor threshold (detail).. Crossover is achieved for all C_{floor} up to \$3.0M (the stochastic upper bound) and persists to \$10M ($N^* = 10,847$) via finite-horizon amortization (the permanent crossover threshold $C_{\text{floor}} < \$1.67\text{M}$ is derived in §4.5).

Ongoing capital maintenance (detail).. Periodic maintenance at $\phi_K \cdot K$ every Δt_K years adds a recurring cost stream:

$$C_{\text{maint}} = \sum_{j=1}^{\lfloor T/\Delta t_K \rfloor} \frac{\phi_K \cdot K}{(1+r)^{j \cdot \Delta t_K}}, \quad (\text{A.1})$$

where $T = N/\dot{n}_{\text{max}}$ is the program duration. With $\Delta t_K = 5$ yr: $\phi_K = 5\%$ shifts the crossover by +1,435; $\phi_K = 10\%$ by +3,215; $\phi_K = 15\%$ by +5,515. The crossover remains within the MC range for $\phi_K \leq 10\%$.

Earth manufacturing cost floor (detail).. To test whether the Wright learning curve overestimates cost reduction at high volumes (where Earth manufacturing may transition to commodity-like production with material-dominated costs), we sweep $C_{\text{mfg}}^{\text{floor}} \in \{0, \$2\text{M}, \$5\text{M}, \$10\text{M}\}$. The crossover is unaffected at all tested floor values up to \$3M ($N^* = 4,077$ throughout). This is because at the crossover volume ($\sim 4,400$ units), the unfloored Earth mfg cost ($C_{\text{mfg}}^{(1)} \cdot n^{b_E} \approx \8M) is already above even the highest floor tested. At $n = 10,000$: with no floor, Earth unit cost is \$10.0M; with $C_{\text{mfg}}^{\text{floor}} = \10M , it rises to \$11.4M. The floor would matter only for production volumes well beyond 10,000 units, where the learning curve would have driven mfg cost below the floor.

Capital–production rate correlation.. The baseline Monte Carlo correlates ISRU capital K and production rate $\dot{n}_{\text{max}}$ ($\rho_{K,\dot{n}} = 0.5$) via the extended Gaussian copula (§3), reflecting that higher-capacity facilities require larger capital investment. This eliminates implausible corner cases (e.g., $K = \$100\text{B}$ with $\dot{n}_{\text{max}} = 250$ units/yr). As a sensitivity test, we run with $\rho_{K,\dot{n}} = 0$ (full independence): the convergence rate decreases from 79.4% to 76.6% and the conditional median moves from 5,800 to 5,615 (−185 units, −3%). These results confirm that the correlation assumption is not a material driver of the conclusions.

Extended 6D copula.. The baseline 3D copula correlates only $(p_{\text{launch}}, K, \dot{n}_{\text{max}})$. To test whether plausible correlations among the remaining parameters materially affect results, we extend the copula to 6D, adding $\rho(K, t_0) = +0.4$ (larger facility $\rightarrow$ longer ramp-up), $\rho(K, A) = -0.3$ (larger facility $\rightarrow$ more downtime), and $\rho(K, C_{\text{ops}}^{(1)}) = +0.3$ (larger facility $\rightarrow$ higher initial ops). All other new off-diagonals are zero; existing correlations are preserved. Table A.24 compares the results at $r = 5\%$.

Table A.24: Extended copula sensitivity: 3D (baseline) vs. 6D copula (10,000 runs, $r = 5\%$, $H = 40,000$, same seed).

Copula	Conv. %	Cond. median	Shift
3D (baseline)	85.1%	4,311	—
6D (extended)	84.6%	4,091	−0.5 pp

The extended copula shifts convergence by -0.5 pp and the conditional median by -220 units ($\sim -5\%$), confirming that correlations among the three additional parameters do not materially alter the headline results. The negative median shift arises because $\rho(K, A) < 0$ penalizes high- K scenarios with lower availability, slightly reducing the number of marginally non-converging runs.

Asymmetric learning plateau.. The symmetric plateau test (§4.2) assumes both pathways plateau simultaneously. A more adverse scenario for ISRU is one in which Earth learning continues unabated while ISRU learning stalls. With Earth damping = 1.0 (no plateau) and ISRU damping = 0.3 at $n_{\text{break}} = 200$: $N^* = 4,541$ (shift: +167). With ISRU damping = 0.5 at $n_{\text{break}} = 500$: $N^* = 4,455$ (shift: +81). Both worst-case asymmetric shifts are well below the $\text{LR}_I = 1.0$ boundary test (+1,588), confirming that asymmetric plateau is not a decisive risk factor.

Pioneering phase (negative learning).. The baseline Wright curve assumes monotonically decreasing costs from unit 1. In practice, the first ISRU units may experience cost increases due to debugging, equipment failures, and process redesign. We test a two-phase model with cost multiplied by $\gamma \geq 1$ for the first n_p units. At $\gamma = 2$, $n_p = 100$: +34 units (+0.8%); $n_p = 500$: +136 (+3.1%); $n_p = 1,000$: +246 (+5.6%). At $\gamma = 5$, $n_p = 500$: +559 (+12.7%); $n_p = 1,000$: +1,037 (+23.6%). The $\text{LR}_I = 1.0$ boundary test (+1,679 units) remains more conservative than any pioneering phase at $n_p \leq 500$.

Quality assurance and certification costs.. We add a QA cost $C_{\text{QA}}^{(1)}$ declining with learning rate LR_{QA} . At $C_{\text{QA}}^{(1)} = \$1\text{M}$, $\text{LR}_{\text{QA}} = 0.85$: +73 units (+1.7%). At $C_{\text{QA}}^{(1)} = \$3\text{M}$, $\text{LR}_{\text{QA}} = 0.90$: +421 units (+9.6%). Even conservative QA assumptions shift the crossover by $< 10\%$.

K clip bound sensitivity (detail)..

Table A.25: K clip upper bound sensitivity (10,000 runs, $r = 5\%$, $H = 40,000$). Raising the clip from \$200B to \$500B has negligible effect on convergence or conditional median.

$K_{\max}$ clip	Conv. %	Cond. median	K P90	K P99
\$200B	85.1%	4,311	\$163B	\$200B
\$300B	84.0%	4,230	\$163B	\$300B
\$500B	83.9%	4,224	\$163B	\$344B

Earth pathway cross-check (detail).. To cross-check the Earth cost model against real production data, we map it to the Iridium NEXT constellation. Thales Alenia Space manufactured 81 satellites (bus mass ~ 860 kg each) under a contract valued at approximately \$2.1B. Setting $C_{\text{mfg}}^{(1)} = \$80\text{M}$, the Wright curve at $\text{LR}_E = 0.80$ predicts cumulative manufacturing cost of \$2.26B at $n = 81$, closely matching the contract value ($2.26/2.1 = 1.07\times$). Reverse-fitting to the \$2.1B total yields an implied $\text{LR}_E \approx 0.79$, consistent with Wertz 2011 [36] (0.83–0.88 for satellite buses, $n = 50\text{--}200$).

Table A.26: $n_0 \times \text{LR}_E$ interaction ($r = 5\%$). Each cell shows N^* ; $>40\text{k}$ indicates no crossover within 40,000 units.

n_0	LR_E			
	0.80	0.85	0.90	0.95
0	21,365	4,403	1,987	1,111
50	25,051	4,600	2,050	1,130
100	27,973	4,720	2,089	1,143
200	35,249	4,907	2,149	1,162
500	$>40\text{k}$	5,309	2,276	1,201

$n_0 \times \text{LR}_E$ interaction (detail)..

Kaplan-Meier survival analysis.. Promoted to main text (Table 16 in §4.3).

Convergence as a function of planning horizon.. The choice of planning horizon $H = 40,000$ units is somewhat arbitrary. Figure A.10 presents the empirical CDF $P(N^* \leq H)$ as a continuous function of H for three discount rates, removing the dependence on any single horizon choice. At $H = 10,000$, convergence is 60% ($r = 5\%$); at $H = 20,000$, 74%; results plateau beyond

$H \approx 30,000$ as marginal scenarios are exhausted. The figure provides a decision tool: a program planner can read off the probability of achieving crossover given their specific production horizon.

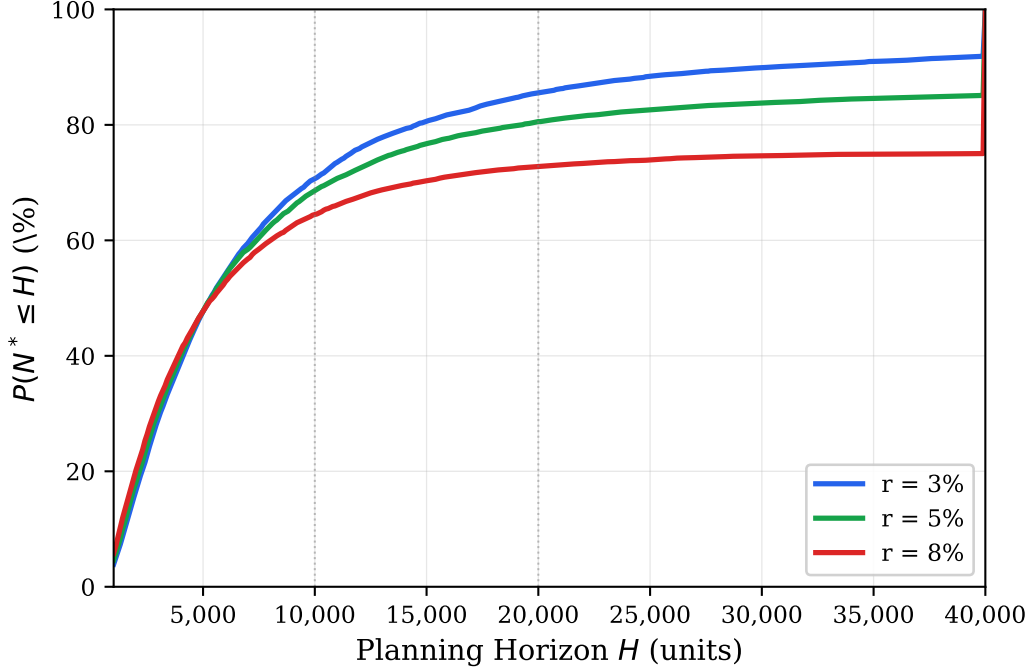


Figure A.10: Probability of achieving crossover $P(N^* \leq H)$ as a continuous function of planning horizon H , for three discount rates ($r = 3\%, 5\%, 8\%$). Based on 10,000 MC runs per rate. Vertical dashed lines mark $H = 10,000$ and $H = 20,000$.

Bootstrap confidence intervals for savings window. The savings window probabilities in Table 19 are computed from 10,000 MC runs. To quantify sampling uncertainty, we construct 95% bootstrap confidence intervals by resampling the converging runs 5,000 times with replacement, computing the savings window probability at each horizon for each resample, and reporting the 2.5th and 97.5th percentiles. The resulting CI widths (± 1.0 – 1.2 pp) confirm that the 10,000-run ensemble provides adequate precision for the headline savings window probabilities (e.g., 95% at 20k units: 95% CI [94%, 95%]).

Appendix B. Schedule and Model Details

Schedule model verification. Substituting $n = 1$ into Eq. 10 yields $t_{1,I} = t_0 + (1/k) \ln(2e^{k/\dot{n}_{\max}} - 1) \approx t_0$, confirming that the first unit is produced near the ramp-up midpoint, with $S(t_0) = 0.50$. The piecewise enforcement ($\dot{n} = 0$ for $t < t_c$) affects only the exponentially small tail before t_c ; $N(t_c)$ is negligible ($\sim 10^{-3}$ units). The counting convention starts from $N(t_0) = 0$, not from facility groundbreaking.

Table B.27: Pathway-specific delivery schedules. The gap column shows the additional time ISRU requires relative to Earth delivery of the same unit number.

Unit n	$t_{n,E}$ (yr)	$t_{n,I}$ (yr)	$S(t_{n,I})$	Gap (yr)
1	0.002	5.00	0.50	+5.00
10	0.02	5.04	0.52	+5.02
100	0.20	5.34	0.66	+5.14
500	1.00	6.31	0.93	+5.31
1,000	2.00	7.34	0.99	+5.34
5,000	10.00	15.35	1.00	+5.35
10,000	20.00	25.35	1.00	+5.35

Deterministic vs. MC capital estimates. The deterministic baseline uses $K = \$50\text{B}$ (the modal estimate for a first-of-kind vertically integrated ISRU facility), while the MC median is $K = \$65\text{B}$. This difference arises because the log-normal distribution is right-skewed: the mode of $\text{LN}(\mu_{\ln}, \sigma_{\ln})$ is $e^{\mu_{\ln} - \sigma_{\ln}^2} = \46B , while the median is $e^{\mu_{\ln}} = \$65\text{B}$. The deterministic $\$50\text{B}$ lies between these values; the MC correctly propagates the full distribution.

Appendix C. Throughput Analysis Details

At 500 Starship launches per year at 100,000 kg payload capacity, Earth can deliver 50,000 tonnes/year. A 10^5 -unit program (185,000 tonnes) would require ~ 3.7 years of dedicated capacity; a 10^6 -unit program would require 18,500 launches over 37 years. Total global launch mass in 2023 was $\sim 1,200$ tonnes [20]. We tested Earth delivery rate caps of 2,000–27,000 units/yr (~ 37 –500 Starship-equivalent launches/yr); at all caps the crossover is unchanged from baseline because the production rate (500 units/yr) is well below even the most conservative cap. Table C.28 maps crossover volumes to illustrative architectures.

Table C.28: Illustrative demand scenarios for 1,850 kg structural modules. Mass estimates are order-of-magnitude [20, 27].

Architecture	Mass	Units	Timeline	Crossover?
1 GW SPS demo*	5,000 t	2,700	10–15 yr	Below N^*
10 GW SPS constellation*	50,000 t	27,000	20–30 yr	Above N^*
O’Neill cylinder	500,000 t	270,000	30–50 yr	Far above N^*

*Revenue-qualified: SPS structural modules may generate $> \$0.9\text{M}/\text{unit}/\text{yr}$, in which case the Earth pathway is preferred on a utility-maximizing basis (§5).

Appendix D. Parameter Justification Details

First-unit ISRU operational cost $C_{\text{ops}}^{(1)} = \$5M$. Sintering or melting regolith requires approximately 1,000 kWh per tonne of processed material [7]; a 1,850 kg unit requiring ~ 5 tonnes of raw feedstock (at $\sim 37\%$ structural yield) consumes $\sim 5,000$ kWh. At lunar surface power costs of $\sim \$100\text{--}200/\text{kWh}$ [30, 22], the energy component is $\$0.5\text{--}1.0M$. Adding equipment wear, consumable replacement, and remote operations overhead yields $\sim \$5M$ under immature conditions.

ISRU capital $K = \$50B$. This covers the complete chain: prospecting, extraction, processing, fabrication, and quality control, deployed without an existing supply chain. For context, ISS lifecycle cost is $\sim \$150B$; Artemis is budgeted at $\sim \$93B$. The capital cost per unit of annual throughput ($K/\dot{n}_{\text{max}} = \$100M$ per unit/year) is comparable to offshore oil platforms ($\$5\text{--}15B$, single-commodity) and nuclear plants ($\$10\text{--}25B$), adjusted for the uniquely challenging operational environment.

K subsystem decomposition.. A rough breakdown of the $\$50B$ baseline:

- *Power system* ($\sim \$8B$): 50 MW nuclear/solar at $\sim \$160M/\text{MW}$.
- *Mining & processing* ($\sim \$12B$): excavation, beneficiation, smelting.
- *Fabrication facility* ($\sim \$10B$): additive manufacturing + assembly.
- *Transport & logistics* ($\sim \$5B$): landing pads, propellant depot, tugs.
- *Habitat & ops support* ($\sim \$5B$): crew/teleops infrastructure.
- *Integration, margin, delivery* ($\sim \$10B$): 20% margin + launch to surface at $\sim \$5,000/\text{kg}$ for ~ 500 t.

These figures are order-of-magnitude estimates for context; the Monte Carlo samples K from the full log-normal distribution rather than decomposing it into subsystems. The K distribution should be interpreted as a scenario range rather than a calibrated estimate. Table 14 shows crossover economics as a function of K median, enabling readers to apply their own capital assumptions.

K-median sensitivity.. Promoted to main text (Table 14 in §4.3).

Earth-sourced component fraction (vitamin model). The vitamin fraction model (Eq. 19 in §3.2.4) modifies the ISRU operational cost to account for components that cannot be ISRU-sourced. Sensitivity tests span $f_v \in \{0, 0.05, 0.10, 0.15, 0.20\}$ and $c_{\text{vit}} \in \{\$5,000, \$10,000, \$50,000\}/\text{kg}$.

Cost basis normalization detail.. The $U[\$100, \$400]$ propellant floor range is grounded in a bottom-up decomposition:

1. *Propellant to LEO:* LOX at $\sim \$0.20/\text{kg}$ and CH_4 at $\sim \$0.40/\text{kg}$, with a propellant-to-payload mass ratio of $\sim 25\text{--}37:1$ for Starship-class vehicles, yields $\sim \$5\text{--}8/\text{kg}$ of payload.
2. *LEO-to-GEO transfer:* $\Delta v \approx 3.9\text{--}4.2 \text{ km/s}$; a chemical tug costs $\sim \$80\text{--}120/\text{kg}$ transferred, or $\sim \$30\text{--}60/\text{kg}$ via solar electric propulsion.
3. *Ground operations:* Pad turnaround, range fees, and GSE amortization at high cadence (>50 flights/yr) yield $\sim \$20\text{--}50/\text{kg}$.

Summing yields $\sim \$105\text{--}178/\text{kg}$ to GEO.

Additional parameter notes. Mass penalty $\alpha \sim U[1.0, 2.0]$ captures additional feedstock required for structural equivalence. Availability $A \sim U[0.70, 0.95]$ accounts for equipment failures and limited maintenance access. Transport cost $p_{\text{transport}} \sim U[\$50, \$300]/\text{kg}$ covers lunar surface-to-GEO delivery [16]. Production rate $\dot{n}_{\text{max}} = 500$ units/year, sampled $U[250, 750]$, requires $\sim 5,000\text{--}10,000$ tonnes/year of raw regolith, within proposed ISRU facility scales [30]. Ricardian rent is not a concern: lunar regolith is abundant ($\sim 10^{15}$ tonnes in the top 5 m) and compositionally homogeneous at the scales modeled.

Appendix E. Assumptions and Limitations

Key simplifying assumptions: (1) fixed unit mass (no design evolution); (2) quality parity between pathways (optimistic for early ISRU); (3) static

Table D.29: Representative vitamin BOM for a 1,850 kg structural module. Components marked [†] are Earth-sourced in the baseline but potentially ISRU-sourced at maturity; only the irreducible mechanical 5% ($f_v = 0.05$) enters the cost model. Sensors/wiring (3%) are not included in f_v because they are separately accounted for in integration overhead; if included, f_v would rise to ~ 0.08 and the crossover would shift by $\sim +200$ units (§4.2).

Component	Source	Mass %	Mass (kg)	Note
Al structure	ISRU	85%	1,573	Regolith-derived alloy
Ti fasteners	Earth	5%	93	Irreducible vitamin (f_v)
Thermal coatings	Earth [†]	4%	74	Potentially ISRU at maturity
Sensors/wiring	Earth	3%	56	Rad-hard electronics [‡]
Seals/gaskets	Earth [†]	3%	56	Polymer, potentially ISRU
Total		100%	1,850	
Irreducible vitamin ($f_v = 0.05$)		5%	93	Modeled fraction

[†]Potentially ISRU-sourced at high technology maturity; modeled as Earth-sourced in baseline.

[‡]Not included in $f_v = 0.05$; accounted for separately in integration overhead.

Table D.30: Active model equations in baseline MC vs. sensitivity variants. “Baseline MC” is the configuration used for the 10,000-run ensemble at each discount rate; sensitivity variants activate or vary additional features one at a time.

Component	Baseline MC	Sensitivity variants
Earth mfg: two-component (Eq. 2)	✓	—
Earth mfg floor (Eq. 3)	off ($C_{\text{mfg}}^{\text{floor}} = 0$)	sweep \$2–10M
Launch learning (Eq. 5)	✓ ($\text{LR}_L = 0.97$)	sweep 0.90–1.00
Vitamin fraction (Eq. 19)	✓ ($f_v = 0.05$)	sweep 0–0.20
ISRU pioneering phase	off ($\gamma = 1$)	$\gamma = 2\text{--}5$, $n_p = 100\text{--}1,000$
ISRU QA cost	off ($C_{\text{QA}}^{(1)} = 0$)	\$1–3M
Piecewise ISRU schedule	✓ ($t_c = t_0 - 1$)	—
Earth ramp-up	off (instant start)	$t_{0,E} = 1\text{--}2$ yr
ISRU availability	✓ ($A \sim U[0.70, 0.95]$)	—
Learning plateau (Earth)	✓ (stochastic)	$n_{\text{break}} = 200\text{--}2,000$, $\eta = 0.3\text{--}0.7$
Learning plateau (ISRU)	off	$n_{\text{break}} = 200\text{--}1,000$, $\eta = 0.5$
Dynamic vitamin fraction	✓ ($f_v^{\text{floor}} \sim U[0.01, 0.03]$)	$n_v \sim U[2000, 10000]$
σ_{ln} sensitivity	0.70	1.0, 1.3
Earth scaling penalty	off ($\gamma_E = 1$)	$\gamma_E = 1.1\text{--}1.3$, $n_{\text{scale}} = 2,000$
Transport duration τ_{trans}	✓ (0.5 yr)	sweep 0.25–2.0 yr

Table D.31: Empirical learning rates from analogous production programs. Our ranges $LR_E \sim \mathcal{N}(0.85, 0.03)$ $[0.75, 0.95]$ and $LR_I \sim \mathcal{N}(0.90, 0.03)$ $[0.80, 0.98]$ encompass these empirical values.

Program class	LR range	n range	Source
Satellite buses	0.83–0.88	50–200	Wertz 2011 [36]
Solar arrays	0.80–0.85	100–1,000	Nagy et al. 2013 [24]
Aircraft structures	0.80–0.85	500+	Argote & Epple 1990 [2]
Metal additive mfg.	0.85–0.92	10–100	Baumers et al. 2016 [4]

production environment (no disruptive innovations); (4) single product type (multi-product would shift crossover earlier); (5) constant real discount rate (risk-adjusted rates differ between pathways in practice); (6) Earth production capacity assumed available from program start (tested in Earth ramp-up robustness check). The phased capital model couples capex spending to the construction schedule: tranches are paid during $[t_0 - 5, t_0]$, ending at commissioning (§4.4). At the deterministic baseline ($t_0 = 5$), coupled and uncoupled phasing are identical; the coupling matters in the MC where $t_0 \sim U[3, 8]$.

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